

Not All Cops Are Bayesian: Selection Neglect and Biased Crime Beliefs

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Abstract

Crime predictions guide patrol allocation, stops, and searches, but crime data is not a representative sample of crime. Because minorities are over-selected into crime records, officers who fail to adjust for data selection may form biased beliefs about crime. We provide experimental evidence of this mechanism in a lab-in-the-field experiment with senior police officers in Colombia. Officers neglect data selection, translating differences in crime observability into prediction bias. Because neglect is a representational failure, algorithmic advice or information provision fail to debias predictions. Selection neglect offers a cognitive foundation for policing disparities.

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I Introduction

We often make predictions based on non-representative samples. For example, we estimate the quality of a product based on the opinion of customers who have self-selected into giving a review. If we don't adjust our predictions for the selection of the sample, selection bias will carry into our predictions: maybe the product isn't that bad, it's just that unhappy customers are more likely to write a review. A growing literature shows that people often fail to account for such selection and treat observed data as if it were representative (Enke, 2020). This cognitive failure, known as selection neglect, is pervasive when forming beliefs about the income distribution, political views, or asset returns (Brundage et al., 2024; Cruces et al., 2013; DellaVigna & Kaplan, 2007; Malmendier & Shanthikumar, 2007). Crucially, it might have particularly relevant consequences in policing, where high-stakes decisions are increasingly driven by selected crime data (Brayne, 2020).

A defining feature of policing is that not all crimes are observed, and crime data is not a representative sample of the underlying crime. A broad consensus shows that minorities are patrolled, stopped, searched, and arrested more often than observably similar white individuals (Aggarwal et al., 2025; Ba et al., 2021; Chen et al., 2023; Feigenberg & Miller, 2025; Gonçalves & Mello, 2021; National Academies of Sciences, Engineering, and Medicine, 2023; Pierson et al., 2020; Vomfell & Stewart, 2021). Due to these enforcement disparities, the crimes of some groups are more likely to be observed and become crime data. In other words, crime data suffer from selection bias, so predictions that don't correct for observability will reflect that same bias ¹. Because data-driven crime predictions guide enforcement decisions —where to patrol, who to stop, or when to file out a field interview, prediction bias can feed back into selection bias. If officers predict minorities to be more likely to commit a crime, they will focus enforcement on them, in turn making crime more observable among that group ². Selection neglect can be a cognitive mechanism for policing disparities by generating cycles of biased beliefs.

Although selection bias is pervasive in crime data, there is no evidence on how officers make crime predictions from such data. There are several reasons for this. First, while crime predictions guide enforcement decisions, the exact mapping remains unknown, making it difficult to back-up predictions from administrative data on enforcement. Second, identifying selection neglect from administrative data would require variation in the observability of crime that is unrelated to enforcement and to underlying crime. In practice, such variation is rare: enforcement decisions and crime observability are often the same, and criminals respond to changes in observability (Gonçalves, Jacome, & Weisburst, 2025). As a result, despite theoretical work has pointed to the role of selection neglect in policing (Che et al., 2024; Hübert & Little, 2023), we lack direct evidence of whether police officers neglect selection, why, and what are the consequences for prediction disparities.

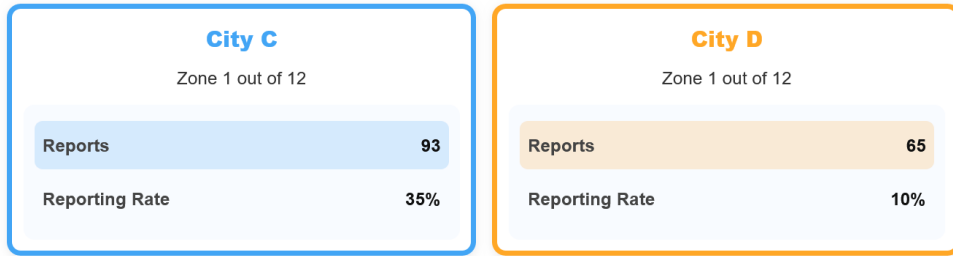
In this paper, we document that police officers neglect data selection and form biased beliefs about crime. In a novel lab-in-the-field experiment, where we rule out racial animus, stereotypes, or institutional frictions, we show that officers carry selection bias into biased predictions, causing disparities in beliefs when crime is more likely to be observed among one group. We present officers with twenty crime

¹Unbiased estimation using police-generated data has proven challenging even for trained researchers (Arnold et al., 2025; Gonçalves, Mello, & Weisburst, 2025; Knox et al., 2020).

²Chen et al. (2023) find that 60% of the racial gap in arrests in the US can be explained by differences in patrolling intensity across neighborhoods.

prediction problems. In each, they see data about two fictitious neighborhoods and are incentivized to correctly predict in which of them more crimes happened. For each neighborhood, officers see the number of *reported crimes* (how many crimes have been observed), as well as the *reporting rate* (the likelihood that a crime is observed). Officers predict crime from a selected sample of reported crimes, and can easily adjust for selection as determined by the reporting rate. Take the example in Figure 1, where 93 crimes were reported in neighborhood C and 65 in D. An officer who doesn’t adjust for selection would estimate more crime happened in C. However, there is selection bias in this sample: while around one in three crimes are reported in C, only one in ten is observed in D. Adjusting by observability, an officer should estimate more crime happened in D (expected 650 crimes) than in C (265 crimes). By generating random variation in observability through the reporting rates while holding underlying crime constant, this simple design allows us to identify whether officers account for selection bias or instead take selected data at face value.

Figure 1: Main Experimental Design Caption



We implement this experiment with senior officers of the National Police of Colombia, many of which have among their responsibilities to predict crime and allocate patrols over space. The National Police uses reported crime as their main crime data³ Thus, crime data selection is determined by reporting behavior, which is transparent to officers and well documented through victimization surveys (DANE, 2023). Because crime data is selected in a transparent way, Colombia is a well suited setting to study neglect, and our experiment includes several features to identify neglect in this setting. First, crime prediction in the experiment is relative, not absolute. We make this design choice because officers are not used to making continuous estimates of crime but relative comparisons. While continuous estimates underlie the prediction, the binary choice is a more familiar environment. Second, we match what determines crime observability in the field: reporting behavior. We use reporting rates as a simple measure of observability that is transparent to officers. Finally, we abstract from stereotypes and animus by using abstract neighborhoods.

We find robust evidence that police officers neglect selection when making predictions from crime data. Our experimental design generates two types of decisions. In cases where the area with more reports is also the area with more crime, so adjusting for selection doesn’t change the prediction, officers make the wrong prediction in less than a fourth of decisions. In contrast, officers systematically make the wrong prediction in cases where adjusting for selection is necessary to identify the area with more crime, as in Figure 1. Because the only difference between these two cases is that selection bias changes the optimal prediction, selection neglect causes a 36 percentage point excess in prediction errors. These

³In Colombia, crime can be reported both by citizens and by police officers and prosecutors. In 2024, over 80% of reports were filed by citizens (CEJ, 2025).

excess error is economically significant: it accounts to roughly half of the average crime gap between areas. This pattern persists through the experiment and is robust to focusing on officers who made no mistake on the instruction comprehension questions or on a numeracy test.

Selection neglect doesn't only increase the magnitude of prediction bias, it also predicts its direction. While our previous results rely on random variation in reporting rates at the neighborhood level, in the field there is selection bias across groups. We introduce groups in our experiment by bundling neighborhoods in two abstract cities, this is, two groups with no distinct social meaning. We then exogenously vary whether observability of crime varies across cities while keeping crime identical. We implement two conditions. In the baseline condition, reporting rates are identically distributed in both cities. In the Selection Bias Across Groups condition, reporting rates are systematically higher in one city, creating selection bias across groups. We find no prediction bias across cities in the baseline condition: officers are equally likely to predict higher crime in both cities. In contrast, officers in the Selection Bias Across Groups condition become 20 percentage points more likely to predict crime in the city where crime is more observable. Thus, selection bias translates into systematic prediction bias across groups, despite crime being identical. These results confirm selection neglect as a micro-foundation for statistical discrimination: biased crime beliefs arise from selected data even in the absence of stereotypes or preferences.

In many field settings, crime data is endogenously generated: enforcement determines which crimes are observed, and enforcement itself is guided by observed data. Neglect can be particularly harmful in such environments, as it might lead to persistent cycles of bias (Che et al., 2024). However, it is possible that officers react differently to exogenous versus endogenous sample selection. We test whether officers neglect endogenous selection by running an additional treatment. In the Endogenous Selection condition, predicting higher crime in one city increases reporting rates in that city for the next decision, but without affecting underlying crime. That is, we induce endogenous variation in observability, while keeping crime constant. We find neglect persists in this condition: endogenous selection increases prediction error by 38 percentage points. As a result, the average officer becomes more likely to predict higher crime among the same group in two consecutive decisions, and 21% of officers do so through all decisions in the condition. Thus, endogenous selection leads to persistent statistical discrimination.

Endogeneity and selection bias across groups often coincide in crime data in the field. We next explore the consequences of neglect in such a selection environment by focusing on the subset of officers who are randomly selected into the Selection Bias Across Groups and Endogenous Selection treatments together. We find that neglect persists in this environment and its consequences compound. Officers in the combined condition are 40 percentage points more likely to predict higher crime in the city where crime is more observable, and 37% of them always predict higher crime there. Taken together, these findings show that neglect interacts with the selection environment, translating selection bias into persistent statistical discrimination.

After exploring the consequences of selection neglect, we turn to its mechanisms. To better know how to mitigate neglect, it is crucial to understand at what margin of the inference process this failure arises. Making the correct crime prediction in this setting involves two margins: (1) being aware of selection when making predictions—including observability in the inference process, and (2) adjusting correctly for selection. Our experimental design decomposes mistakes into these two margins. When given the

option to view reporting rates or not before making a prediction, 36% of officers choose to ignore the measure of observability. This deliberate inattention suggests that selection isn't part of the inference problem at all for over a third of officers. Despite its large magnitude, failures at this margin don't explain the totality of selection neglect: among the two thirds of officers who choose to view reporting rates, only 18% get all predictions right. Most officers incorporate selection into their predictions but fail to correctly adjust for it.

What drives selection neglect on these two margins? Our evidence points to flawed mental models, rather than salience or computational frictions. At the first margin, where selection fails to come to mind at all, neglect is related to experience: officers with more practice predicting from selected crime data are significantly more likely to look at reporting rates. Likewise, officers who are aware of selection bias in the crime data they use in the field show no neglect at all. In contrast, simply making reporting rates more salient in the experiment does nothing to reduce neglect, and we find limited evidence of computational frictions.

To further explore the mechanisms behind neglect, we run two additional treatments. Officers either: (i) receive a recommendation from an algorithm that neglects reporting rates, (ii) receive a recommendation from an algorithm that correctly accounts for selection, or (iii) get no recommendation at all. Officers are explicitly told which algorithm they were assigned and are free to follow or override its advice. We find officers mostly override recommendations that clash with the inference rule they would have applied in the absence of the algorithm. As a result, even an algorithm that correctly predicts crime fails to improve their predictions or reduce neglect. This pattern rules out salience or computational frictions as the key drivers of neglect: the recommendations are highly salient and require zero calculation. Instead, the evidence points to a deeper problem: neglect is, at its core, a failure of representation. Interventions that provide better predictions or more information about selection will likely be ineffective if they fail to change officers' mental model about how crime data is generated.

Our findings provide direct belief-level evidence that selection bias in crime data generates statistical discrimination even in the absence of stereotypes or racial animus. While theoretical work had suggested this source of disparities (Che et al., 2024; Hübert & Little, 2023), which also appears in predictive algorithms (Arnold et al., 2025; Lum & Isaac, 2016), direct evidence from police officers was missing. This paper provides this evidence, building on an extensive literature on policing disparities consistent with statistical discrimination (Ba et al., 2021; Chen et al., 2023; Feigenberg & Miller, 2025; Gonçalves & Mello, 2021; Hoekstra & Sloan, 2022; Pierson et al., 2020; Vomfell & Stewart, 2021). We contribute to this agenda by directly measuring how police officers form beliefs from selected data. Our findings support that cognitive failures and not only preferences might explain policing inefficiencies. This paper thus joins a recent but growing literature focused on cognitive and emotional mechanisms of officer decision-making, from the role of cognitive reflection (Dube et al., 2024; Owens et al., 2018) and cognitive depletion (Ferrazares, 2025) to emotional responses to peer risks (Holz et al., 2023). More broadly, it contributes to research studying how agents in the criminal justice system form beliefs (Angelova et al., 2025; Arnold et al., 2018; Bhuller & Sigstad, 2024).

We also contribute to the literature studying how individuals draw inference from selected data. Recent studies show that individuals often fail to account for data selection in laboratory settings (Ali et al., 2021; Araujo et al., 2021; Barron et al., 2024; Enke, 2020; Esponda & Vespa, 2018; Farina

& Herman, 2025; Jin et al., 2021). However, it remains unclear whether this failure is also present among field practitioners who routinely work with selected data, as prior work has found inconsistent results. For instance, while Malmendier and Shanthikumar (2007) find that large traders discount selected recommendations more than small traders, Barrios et al. (2025) show that laypeople discount conflict of interest in research more than trained economists, and Koehler and Mercer (2009) document neglect even among financial professionals. Using a lab-in-the-field experiment with senior police officers, we show that selection neglect persists among trained professionals, causing large and consequential prediction errors.

Studying a sample of practitioners also allows us to leverage heterogeneity in their awareness of selection bias in real data, which is difficult to do with experimental subjects doing abstract tasks. We find that awareness of selection in the crime data police officers work with in the field negatively predicts selection neglect in our experiment, suggesting that neglect arises from a deeply rooted representational failure. Prior work has shown that mental representations of inference problems affect what information is demanded and how it is used (Bordalo et al., 2026; Esponda et al., 2024). Our paper extends these findings to practitioners in the field. However, practitioners develop stable representations of the problem they routinely face, in contrast to experimental subjects who make ad-hoc representations. We find that information provision, algorithmic advice, or salience boosts fail to reduce neglect, likely because they fail to change the established mental models practitioners carry to the lab. Thus, we contribute to the experimental literature by showing why interventions that improve decisions in the lab might fail to replicate with practitioners on the field. These findings connect with prior work showing that decision makers often override algorithmic advice when it conflicts with their established mental model, making algorithms ineffective (Angelova et al., 2025; Hoffman et al., 2018; Stevenson & Doleac, 2024).

The rest of the paper proceeds as follows. Section II introduces the setting of policing in Colombia and our sample of police officers. Section III describes the experimental design and the empirical strategy. Section IV presents the results on the prevalence of selection neglect, as well as its consequences for disparities. Section V explores the mechanisms driving neglect, using algorithmic advice to tell them apart. Section VI discusses related literature and concludes.

II Setting

A Data-Driven Policing in Colombia

Policing in Colombia is explicitly data-driven. Station commanders and operational officers routinely rely on administrative crime data to form beliefs about where and when crime is most likely to occur, and to respond accordingly through patrol allocation and other operational decisions. While objectives may differ across officers and contexts, accurate beliefs about the spatial and temporal distribution of crime are a necessary input to any policing strategy. The primary source of information the National Police of Colombia uses for these crime predictions is reported crime data. Internal monitoring systems aggregate citizen crime reports across locations and time, and officers are trained to interpret variation in reported crime as informative about underlying criminal activity. In practice, reported crime counts are the dominant metric used to summarize crime patterns within an officer’s area of responsibility.

A central challenge, however, is that reported crime data is not a representative sample of crime. Crimes enter administrative records only if they are reported by citizens, and the probability that a

crime is reported varies substantially across places, crime types, and populations. According to national victimization surveys, only around 31% of crimes in Colombia are ever reported to the police (DANE, 2023). Reporting rates are high for some offenses, such as homicide, but extremely low for others, such as extortion; they also differ markedly across urban and rural areas, and across the income distribution⁴. As a result, two locations with identical underlying crime levels can generate very different volumes of reported crime solely because crimes are observed with different probabilities.

This reporting-based selection creates a selection problem for belief formation. If officers treat reported crime counts as a representative measure of underlying crime, variation in reporting behavior will be mechanically confounded with variation in criminal activity. A rational decision-maker would therefore need to adjust reported crime data for the probability that crimes are reported in order to form accurate beliefs about true crime levels. One might argue that officers fail to adjust reported crime for selection simply because they lack sufficiently granular information about reporting behavior. While officers may not observe reporting rates at fine geographic levels, coarse information about selection is available in victimization surveys, and yet this knowledge is rarely operationalized in practice. Our experimental design addresses this concern directly by making reporting rates fully observable, allowing us to test whether officers adjust for selection when informational constraints are removed.

Importantly for our purposes, the selection problem of crime data is both transparent and plausibly exogenous to individual officers’ short-run decisions. In the Colombian context, selection into crime data is driven primarily by citizen reporting behavior rather than by discretionary police actions such as stops or arrests⁵. While policing may affect reporting over longer horizons through trust and visibility, officers do not directly control whether specific crimes are reported at the moment the data are generated. These features make Colombia an unusually well-suited setting to study belief formation from selected data. Officers routinely rely on reported crime data to form beliefs; the selection process that generates this data is simple, salient, and understandable; and selection is not mechanically tied to officers’ own enforcement actions in the short run. This combination allows us to isolate whether expert decision-makers correctly adjust for known data selection while abstracting from preferences, stereotypes, and institutional constraints that confound inference in observational settings.

B Sample

Our participants are Captains of the National Police of Colombia, the rank that typically serves as station commander and is directly responsible for interpreting crime data and forming crime predictions that guide patrol allocation. We focus on captains because they are the officers who routinely work with reported crime data in practice and whose beliefs about crime patterns are operationally relevant.

Logistics. — In September 2025, 202 captains of the National Police of Colombia were attending the Police Postgraduate School (ESPOL) in Bogotá, completing the mandatory training required for

⁴National victimization surveys uncover systematic variation in reporting rates. The ECSC survey by DANE (2023) documents that while 43% of crimes are reported in Bogotá, only 22% are in rural areas. Within Bogotá, citizens with higher socioeconomic status (strata V and VI) are twice as likely to report a crime than lower status ones (strata I and II). In terms of types of crime, while less than 25% of violence and extortion cases are reported, around half of vehicle thefts are reported, and most homicides are.

⁵See Gonçalves, Jacome, and Weisburst (2025) for how officer behavior determines selection into crime data in the US context.

promotion to Major (NATO OF-4). These officers constitute the universe of captains eligible for promotion in that cohort and were instructed by superior officers to attend our experimental session as part of their training activities. Upon arrival, officers were invited to participate in a “Policing Science Study” and were offered an incentive of USD 5 to 20. Eighteen officers (8.9%) declined to participate before receiving any information about the experiment. Of the 184 officers who began the study, all but two completed it, leaving a final sample of 182 participants and no evidence of endogenous attrition.

Final sample. — The final sample consists of 182 captains with substantial professional experience. Officers have an average of 14 years of service in the National Police of Colombia (minimum 12 years), and all hold at least an undergraduate degree, a requirement for promotion to this rank; 15% also hold a master’s degree. Importantly for our purposes, participants have extensive experience making operational decisions: 25% were serving as station commanders immediately prior to the training, and an additional 40% held operative positions involving the allocation of police resources. Because our sample consists of captains eligible for promotion, it is, if anything, positively selected in terms of ability and experience. In terms of demographics, 83% of participants are men, and ages range from 32 to 45 years (mean 36). Overall, the sample comprises experienced practitioners who routinely interpret crime data in the field, making them a particularly appropriate population to study belief formation from selected data.

III Design

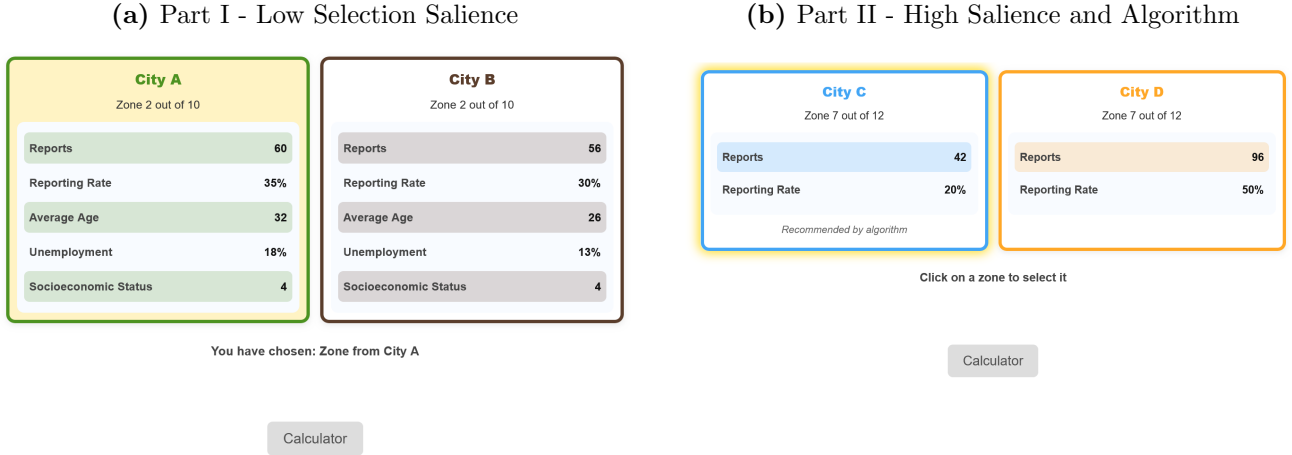
Our experimental design isolates the belief-formation problem that police officers face when interpreting crime data, abstracting from enforcement choices, institutional constraints, and incentives. In practice, policing decisions depend on a variety of considerations, including objectives, resources, and strategic interactions. Rather than attempting to replicate this complexity, our design focuses on a more fundamental input to those decisions: officers’ beliefs about where crime is more likely to occur. We therefore study crime prediction as a cognitive task in its own right. In the experiment, officers are repeatedly asked to predict which of two locations experienced more crime based on available data. These predictions are not treated as enforcement decisions and do not carry operational consequences. Instead, they are interpreted as discrete elicited beliefs about relative crime levels, analogous to the implicit predictions officers form when reading crime reports in the field. This abstraction is deliberate. By separating belief formation from action, we are able to cleanly identify whether officers correctly account for data selection when forming predictions. Throughout the paper, we therefore interpret experimental choices as measures of crime prediction accuracy rather than as proxies for real-world policing behavior.

A Crime Prediction Task

The Crime Prediction Task is the core belief elicitation mechanism in the experiment. Participants see the *number of crime reports* from two fictitious neighborhoods and guess which of the two has a higher level of crime. The ground-truth levels of crime are unobservable to participants and participants receive \$5 if the prediction randomly chosen for payment is correct. Alongside the number of crime reports, participants see the *reporting rate* of the neighborhood, this is, the likelihood that a crime is reported. In this context, the reporting rate captures the observability of crime and determines the selection of crime events into crime data. Panel (b) of Figure 2 shows an example of such crime predictions. The left neighborhood received 42 crime reports, while the right one received 96 reports. An officer that neglects

data selection would predict higher crime in the right one, as more crimes were reported there. In contrast, the optimal prediction involves taking into account the relative observability of crime across areas: only 20% of crimes are reported in the left neighborhood, whereas 50% are in the right one. Once the selection of crime reports is taken into account, the expected number of crimes is higher in the left neighborhood⁶. We randomize the true level of crime and the reporting rate across neighborhoods, generating random variation in the selection bias across pairs of neighborhoods. This simple binary prediction allows us to identify whether officers adjust crime data for selection or instead take crime reports at face value.

Figure 2: Crime Prediction Task Screens



Notes: Panel (a) shows the decision screen for Part I, in an example where the participant has guessed the neighborhood from City A to have more crime. Note that in Part I, reports and reporting rates are accompanied by demographic decoys, and that they are always presented as the first data sources. This makes it more difficult to neglect selection. Panel (b) shows the decision screen for Part II, in an example where the participant has not yet made a prediction but sees the recommendation of the algorithm as a golden halo around the zone and a small span in the bottom. Algorithm predictions, as detailed below, are only provided for the last six decisions of Part II and for a subset of participants. Without the recommendation, the screen is identical but without the halo and the text at bottom. Note the on-screen calculator available throughout the experiment.

Throughout the experiment, participants repeatedly make crime predictions about pairs of neighborhoods. We group neighborhoods in abstract cities: the left side of the pairs always belongs to one city and the right one to another. For instance, Panel (b) of Figure 2 shows that the left neighborhood belongs to City C, whereas the right one belongs to City D. Each city consists of a continuum of neighborhoods that share the same distribution of crime and reporting rates. While participants are aware that average crime might differ across cities, they never observe the crime distribution of a city, and thus should have no priors nor stereotypes. City membership is made salient to participants by randomly assigning a color to each city, and showing this color and a city label in each neighborhood. Note that cities serve as abstract labels for any groups with potentially different crime distributions (e.g., age, socio-economic status or race). By using abstract cities rather than socially salient groups, the design rules out taste-based discrimination and isolates inference from selected data. We later use cities to study whether selection neglect creates statistical discrimination across groups.

Information available. — In particular, participants observe fictitious data for two cities $g \in \{A, B\}$,

⁶The expected number of crimes in the left neighborhood is $42/0.2 = 210$, while in the right one is $96/0.5 = 192$.

each consisting of a continuum of neighborhoods. In each neighborhood gn , a number of crimes c_{gn} occurs, which is unobservable to the participant. In every decision, officers observe a pair of neighborhoods (A_n, B_n) . To form predictions about the latent crime level c_{gn} , participants observe crime data. Specifically, they see the number of crime reports r_{gn} and the reporting rate s_{gn} of each neighborhood, defined as the probability that a crime is reported. Two neighborhoods with identical crime levels can therefore generate different numbers of reports solely because of differences in reporting rates. Participants see this information and are incentivized to guess the neighborhood with the higher expected number of crimes: $\max_g E[c_{gn}|r_{gn}, s_{gn}]$

In the first part of the experiment, participants also observe demographic characteristics $\mathbf{X}_{gn}$ —average age, unemployment rate, and socio-economic status (SES).⁷ These demographic variables serve two purposes. First, they act as decoys that reduce the salience of the reporting rate. Second, if officers rely on demographics in a setting where crime data are fully informative, their choices provide a measure of the extent to which non-crime information enters routine crime predictions.

Data-generating process. — Crime in each neighborhood is independently drawn from a city-specific distribution, $c_{gn} \sim \mathcal{N}(\mu_g, \sigma^2)$. Participants don’t know these distributions. Reporting rates s_{gn} are drawn independently from a uniform distribution $\mathcal{U}[0.1, 0.9]$ and rounded to the nearest multiple of five to ease computation. Each crime is independently either reported or not, and the probability of being reported is the reporting rate of the neighborhood s_{gn} . Therefore, the number of crime reports can be described as a binomial variable $r_{gn} \sim \text{Bin}(c_{gn}, s_{gn})$, with $E[r_{gn}] = c_{gn}s_{gn}$. Reported crime is therefore a *selected* signal of underlying crime, as it depends on the observability of crime given by the reporting rate s_{gn} . Conditional on the number of crime reports and the reporting rate, the optimal prediction adjusts for selection by scaling reports by the reporting rate: $E[c_{gn}|r_{gn}, s_{gn}] = r_{gn}/s_{gn}$. To isolate inference errors from statistical noise, instead of actually drawing the number of crime reports r_{gn} from a binomial distribution we fix it to its expected value, setting $r_{gn} = c_{gn}s_{gn}$. Under this design, a decision-maker who correctly adjusts for selection always makes the correct prediction. Any deviation from the optimal rule therefore reflects cognitive error rather than sampling variation. Demographic variables $\mathbf{X}_{gn}$ are drawn independently from uniform distributions and are, by construction, uninformative about crime.

What participants know about the DGP. — Participants are informed that each neighborhood has an uncertain level of crime c_{gn} and that reported crime depends on reporting behavior. They are explicitly told that the reporting rate s_{gn} represents the probability that each crime is reported, so that in expectation $E[r_{gn}] = c_{gn}s_{gn}$. To ensure comprehension, participants answer questions that require applying this logic⁸. Crucially, participants learn that reporting rates are independent of the underlying number of crimes and the demographic variables. By making the selection process fully observable, the design rules out informational constraints about the data-generating process and allows us to test whether officers adjust for selection when the relevant information is available.

⁷In Colombia, residential areas are classified into socio-economic strata (*estrato*) ranging from 1 (low) to 6 (high). This classification is used to set utility prices and target public services and is a familiar proxy for income for police officers.

⁸Instructions include examples such as: “Imagine that 200 crimes occurred in a neighborhood. If the reporting rate is 50%, we expect 100 crimes to be reported.”

B Experiment Structure

Introduction — After listening to the instructions, participants complete a brief training and answer several comprehension questions designed to verify their understanding of the data-generating process, including the relationship between reported crime, reporting rates, and underlying crime. Incorrect answers are corrected before participants proceed. Participants also answer some incentivized arithmetic questions involving the basic operations required to adjust reported crime for selection. These steps allow us to assess whether observed prediction errors reflect task misunderstanding or limited numeracy rather than selection neglect. Before starting the task, participants can ask any clarifying questions.

Part I — The purpose of Part I is to elicit crime predictions in an environment where data selection is present but not salient. Officers repeatedly predict which of two neighborhoods experienced more crime while observing reported crime data alongside additional information that does not affect the optimal prediction. This setting mirrors real-world contexts in which reporting rates are available but embedded among multiple variables rather than highlighted.

Part I consists of ten crime predictions across neighborhoods of cities A and B. In the first decision, officers observe only the number of reported crimes in each neighborhood. With no additional information, the best guess is to predict higher crime in the neighborhood with higher reports. We use this first decision as a comprehension check, and drop from the analysis the 16% of participants who don't pass it. In the next eight decisions, officers observe reported crimes, reporting rates, and demographic characteristics. The demographic variables —average age, unemployment, and socio-economic status— are intentionally uninformative about crime and serve to reduce the salience of the reporting rate. This allows us to observe whether officers incorporate data selection into their predictions when it is not made focal. In the final decision of Part I, officers choose which additional variable —out of the reporting rate and the demographic variables— to observe alongside reported crime before making their prediction. Selecting the reporting rate reveals that data selection enters the officer's mental model of the inference problem, while selecting any other variable indicates that selection is not treated as relevant for prediction. This choice provides a direct measure of whether officers consider selection when interpreting crime data in a low-salience environment.

Participants have 1 minute to make each prediction in Part I. To ease computations, officers can use a calculator on screen. We randomly select one prediction of Part I for payment, and pay \$5 if officers correctly guessed the neighborhood with the higher number of crimes.

Part II — Part II builds on the core prediction task by increasing the salience of data selection and studying its implications in more structured environments. To avoid learning or carryover from Part I, all decisions in Part II involve neighborhoods drawn from two new cities, *C* and *D*. Both cities have the same crime distribution. Officers make twelve crime predictions, each comparing a pair of neighborhoods from these cities. Throughout Part II, participants observe only the number of reported crimes and the reporting rate for each neighborhood; no demographic information is shown. By removing decoy information, this environment makes the selection information focal. We use this contrast to test whether salience affects neglect. As in Part I, participants have one minute to make each prediction, may use an on-screen calculator, and one guess is randomly selected for a \$5 payment if correct.

We divide the twelve predictions of Part II in halves. The first half provides the cleanest environment

of the experiment to test for selection neglect: participants have already some experience with the task, and only observe the number of crime reports and reporting rates. We focus our main analyses of selection neglect on this first half. In the second half, we introduce additional treatments to test the effect of endogenous data selection and algorithmic advice.

Across all Part II, we study whether systematic differences in data selection across groups translate into biased crime predictions. Participants are randomly assigned to one of two selection environments. In the baseline Uniform Selection condition, reporting rates for cities C and D are independently drawn from the same distribution, as in Part I. In the Selection Bias Across Groups condition, one city has systematically higher reporting rates than the other: $s_{Cn} \sim \mathcal{U}[0.1, 0.6]$ and $s_{Dn} \sim \mathcal{U}[0.4, 0.9]$, or vice versa, generating a 30 percentage point reporting gap in expectation. Participants are not informed of this systematic difference in observability directly, but can infer it from the available reporting rates. Because both cities have the same distribution of crime, any systematic bias in crime predictions can be attributed to selection neglect rather than accurate inference. We use this treatment to study whether selection neglect can generate inaccurate statistical discrimination across groups when selection is biased across groups.

We then examine how selection neglect interacts with endogenous data generation. For the last six decisions of Part II, participants are further randomized into the baseline Exogenous Selection or an Endogenous Selection condition. In the exogenous condition, reporting rates are drawn independently of past predictions. In the endogenous condition, officers' previous predictions affect future observability: if an officer predicts higher crime in city g in decision $n - 1$, the reporting rate in that city increases by 15 percentage points in decision n , $s'_{gn} = s_{gn} + 0.15$. This captures a key feature of real-world data environments, where crime beliefs and enforcement influence which events are subsequently observed (Ang et al., 2025; Chen et al., 2023; Gonçalves, Jacome, & Weisburst, 2025). Notice that this treatment will have no effect if officers account for selection, but will generate belief persistence if they neglect it.

Finally, we test whether algorithmic recommendations mitigate or amplify inference errors. For the last six decisions of Part II, participants are randomly assigned to one of three conditions: no algorithm (Control), a Correcting Algorithm, or a Neglecting Algorithm. Both algorithms use reported crime data to recommend which neighborhood likely experienced more crime, with some noise. The correcting algorithm adjusts for reporting rates, while the neglecting algorithm relies only on raw reports. Participants are informed of the type of algorithm they receive and may choose to follow or override its recommendation. This design allows us to test whether algorithmic correction compensates for selection neglect, or whether neglect extends to the rejection of corrective advice.

End Questions — After completing the experiment, participants answer a short, unincentivized survey. We ask officers to assess their own performance and whether they would like to receive feedback on how to improve their crime predictions. Crucially, officers report their beliefs about the reliability of reported crime data, including whether crime reports capture spatial and temporal crime patterns, whether they are the best available data, whether citizens know how to report crimes, and whether reported data is generally biased. We follow Kling et al. (2007) and aggregate z-scores of each response into an index of *reported data reliability*, where higher values reflect stronger beliefs that reported crime data faithfully represent true crime. This measure allows us to relate experimental inference errors to officers' real-world beliefs about crime data. Participants also report their years of experience in the National Po-

lice of Colombia and the nature of their most recent position (administrative, operative, or surveillance). Officers in surveillance positions are those acting as station commanders and directly assigning patrols. Finally, participants provide basic demographic information.

Table 1: Summary of Experimental Design

Part I	Ten crime predictions
<i>Decision 1</i>	Only see number of reports (sanity check)
<i>Decisions 2-9</i>	See number of reports, reporting rate, demographics (low salience of selection)
<i>Decision 10</i>	See number of reports and one data of their choice (information choice)
Part II	Twelve crime predictions
	Only see number of reports and reporting rate (high salience of selection)
<i>First Half (1-6)</i>	Used for main analysis, selection bias across groups treatment
<i>Second Half (7-12)</i>	Selection bias across groups and endogenous selection treatment; algorithm
End	Performance guess, trust in reported data, experience, and demographics

Why an Experiment

Identifying selection neglect in policing using administrative data is difficult because the object of interest—how officers form beliefs from crime data—is not directly observed. Doing so would require exogenous variation in the selection of crime data that is unrelated to officer behavior and to true crime, so that changes in decisions could be cleanly attributed to inference rather than to incentives or constraints. In practice, such variation is unlikely to exist. Selection into crime data is often determined by policing itself through patrols, stops, and arrests, and even when driven by citizen reporting, reporting behavior responds to police presence and trust (Ang et al., 2025). Moreover, changes in observability typically affect criminal behavior, so shocks to selection do not leave crime unchanged (Gonçalves, Jacome, & Weisburst, 2025). As a result, observational data cannot disentangle how officers interpret selected crime data from the behavioral and institutional processes that generate it. These challenges motivate an experimental approach. By exogenously varying data selection, holding true crime constant, and eliciting crime predictions directly, the experiment isolates belief formation from enforcement behavior and allows us to identify selection neglect in crime prediction.

C Empirical Strategy

Our analysis follows directly from the experimental design. For each officer and decision, we observe which neighborhood gn they predict to have higher crime. For every decision, there is a unique accurate prediction: choosing the neighborhood with the higher number of crimes c_{gn} . Although officers do not observe c_{gn} , the optimal prediction rule—ranking neighborhoods by adjusted reports $\max_g r_{gn}/s_{gn}$ —is always correct because we eliminate sampling noise from the reporting process. The optimal prediction therefore does not depend on priors or assumptions about the underlying crime distributions. Throughout the analysis, the unit of observation is the officer–decision pair. Because predictions are binary, we summarize decision-relevant information using gaps between neighborhoods, $\Delta(x_n) \equiv x_{An} - x_{Bn}$.

Our empirical strategy focuses on whether officers adjust for or neglect data selection when forming crime predictions. In this setting, neglecting selection corresponds to ranking neighborhoods by reported

crimes r_{gn} rather than by selection-adjusted reports r_{gn}/s_{gn} . Because reporting rates are independently drawn across neighborhoods, ignoring selection does not always change the optimal ranking. In *selection-neutral decisions* the neighborhood with more reports is also the neighborhood with more crimes. In these cases, even an officer who ignores reporting rates makes the correct prediction. We therefore use selection-neutral decisions as a benchmark that captures inference errors unrelated to selection neglect. In contrast, in *selection-sensitive decisions* adjusting for selection reverses the ranking of neighborhoods. In these cases, neglecting selection mechanically leads to an incorrect prediction. Selection-neutral and selection-sensitive decisions are otherwise comparable in structure; they differ only in whether accounting for selection affects the optimal choice.⁹ The difference in prediction accuracy between selection-neutral and selection-sensitive decisions therefore identifies prediction errors attributable to selection neglect.

While accuracy provides a natural measure of inference quality, it abstracts from the magnitude of officers' mistakes. If officers were only neglecting selection in cases where the underlying crime gap is minimal, a large accuracy loss would not necessarily imply large prediction errors. To capture the relevance of selection neglect, we define *missed crimes* as the difference in crimes between the optimal and the actual guess. This measure quantifies how much crime is mispredicted due to selection neglect, not merely whether a prediction is correct.

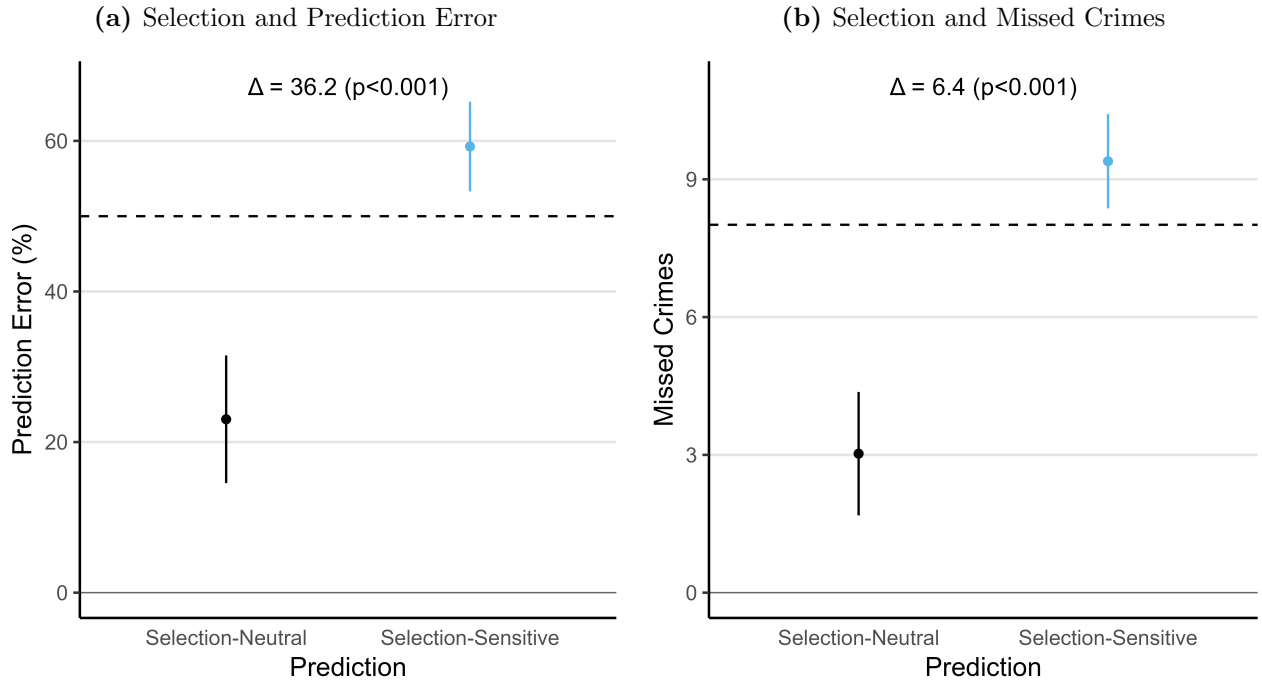
⁹Formally, we denote selection-neutral decisions those that satisfy $\Delta(r_n)\Delta(r_n/s_n) > 0$, while selection-sensitive decisions satisfy $\Delta(r_n)\Delta(r_n/s_n) < 0$. This distinction is not apparent in the data participants observe unless they do the calculations to adjust for selection. For example, if $r_A = 30$ and $r_B = 60$, then with reporting rates $s_A = 21\%$ and $s_B = 40\%$ both raw and adjusted reports imply higher crime in B , making this a selection-neutral decision. With $s_A = 19\%$ and $s_B = 40\%$, raw reports still favor B , but adjusted reports imply higher crime in A , making this a selection-sensitive decision.

IV Selection Neglect and Statistical Discrimination

A Prevalence of Selection Neglect in Crime Predictions

We begin by quantifying how much selection neglect increases prediction errors. Panel (a) of Figure 3 shows that in decisions where neglecting selection does not change the optimal prediction (what we denote *selection-neutral* decisions), officers make the wrong guess in 23% of cases. This error rate already absorbs any mistakes due to inattention or confusion. In contrast, prediction error rises to 59% when the correct prediction requires adjusting for selection (*selection-sensitive* decisions). Thus, neglecting selection causes a 36 percentage point excess in prediction errors. Table A.1 regresses errors on whether the predictions were selection-neutral or selection-sensitive, and finds the selection-driven excess prediction error is highly significant ($p < 0.001$) and robust through the experiment.

Figure 3: Accuracy Loss Due to Selection Neglect



Notes: Panel (a) shows the average prediction error by prediction type. Panel (b) shows the average number of missed crimes —the difference in crimes between the optimal prediction and the actual one, by prediction type. Missed crimes can be interpreted as the average crime gap multiplied by the average error rate. Both panels divide the data by whether adjusting for selection was needed to make the accurate prediction. Selection-neutral predictions, where adjusting for selection doesn't change the optimal prediction, are depicted in black and in the left column; selection-sensitive predictions, where adjusting for selection does change the optimal prediction, are depicted in blue and in the right column. Error bars in both panels represent the 95% confidence interval of the mean, with standard errors clustered at the officer level. The annotations show the difference across types of predictions, alongside with the p-value from a t-test clustered at the officer level. The horizontal solid lines at 0 represent the optimal benchmark, while the dashed lines show the random benchmark.

We next quantify the cost of the prediction error selection neglect induces. We examine how many crimes officers miss relative to the optimal prediction, this is, the crime gap between neighborhoods in inaccurate predictions. *Missed crimes* translate the accuracy loss documented above into a metric directly relevant for policing: how much crime goes unaccounted for because of inference error. While this mapping might seem mechanical, it depends on whether officers just neglect selection in close calls,

this is, in decisions with a small crime gap between neighborhoods. If this was the case, a significant excess prediction error would not translate into a relevant increase in missed crimes. The evidence shows the opposite: selection neglect produces materially worse crime predictions.

Panel (b) of Figure 3 finds that, while officers only miss 3 crimes on average in predictions where adjusting for selection is not needed, this number rises to 9.4 crimes in selection-sensitive decisions. On average, officers miss 6.4 additional crimes ($p < 0.001$) due to selection neglect, representing roughly 40% of the average crime gap. Figure A.3 provides further insight on why neglected crimes increase due to selection: while selection-neutral predictions become easier as the underlying crime gap increases, accuracy remains flat in selection-sensitive predictions. As a result, the excess prediction error due to selection neglect is highest in precisely the situations where errors carry the highest cost in terms of missed crimes.

Selection neglect doesn't only increase the frequency of prediction errors, it also predicts their direction: officers who neglect selection interpret higher observability as higher crime. As a result, officers will predict higher crime in areas where crime is more observable, even when true crime is the same. We test this implication by regressing officers' predictions on differences in reporting rates, conditional on the true crime gap. Table A.3 documents the results of this linear regression. Officers are significantly more likely to predict higher crime in neighborhoods with higher reporting rates: a 1 percentage point increase in the reporting-rate gap increases the guess probability by 0.3 percentage points. This pattern holds even when true crime is held constant, persists across both parts of the experiment, and is robust to including officer fixed effects. These results show that selection neglect mechanically maps selection bias in crime data into biased beliefs.

Robustness

One might be concerned that prediction errors are not driven by selection neglect *per se*, but by officers avoiding a costly calculation in selection-sensitive decisions. We find no evidence to support this. First, the only way to distinguish between neutral and sensitive decisions is computing the selection adjustment, as these decisions are visually identical. Second, officers spend more time in selection-sensitive decisions, but this difference is small (0.07 SD) and not significant ($p = 0.437$). Third, we only find a stronger excess prediction error when restricting the sample to those who spend on average more time on selection-sensitive decisions. Overall, our results can't be driven by less effort in selection-sensitive decisions.

Selection neglect is not explained by confusion about the task or by limited math skills either. We regress again prediction error on whether the decision is selection-sensitive, now adding an interaction with whether the officer made no math mistakes or no instruction errors. Table A.5 finds that selection neglect is robust among officers who answered all nine instruction comprehension questions correctly, as well as those who correctly answered all math questions. This robustness holds both for selection neglect measured as excess prediction error and for missed crimes. Although there is some evidence that officers who lack the necessary math skills show more neglect, officers who have these skills still fail to adjust for selection. In other words, selection neglect arises despite understanding the task and having the basic arithmetic needed to solve it.

Our results are not driven by a handful of officers. Instead, selection neglect is a pervasive failure that

affects the majority of officers. Figure A.4 shows the distribution of errors at the officer level, divided by selection-neutral and sensitive decisions. While the modal officer makes no mistakes when adjusting for selection isn’t needed, they fail all predictions that are selection-sensitive. Focusing on the median officer reveals a similar picture: no mistakes in selection-neutral predictions, but wrong predictions in two thirds of sensitive ones.

B Consequences of Selection Neglect

We have used random variation in crime observability to show that neighborhood-level differences in data selection translate directly into prediction error, as officers treat observed crime as representative of true crime. In practice, selection doesn’t vary randomly: there are systematic differences in observability across groups. We introduce groups in the experiment by bundling neighborhoods—the individual level of inference—in abstract *cities*. Neighborhoods within one city share the same distribution of crime and reporting rates, with each neighborhood being a random draw from the city-level distribution. Each neighborhood in Part II belongs to either City C or City D, with group identification visible through a label and a consistent city color (see Figure 1). Using abstract groups, a standard practice in the literature (Bordalo et al., 2016; Esponda et al., 2023), rules out preferences, stereotypes, or priors about group-level crime.

In this section, we first study how crime predictions respond to selection bias across groups by introducing a treatment where observability differs across cities while keeping crime constant. Then, we turn to the effects of endogenous selection by introducing a treatment where crime predictions affect subsequent observability. While these effects might seem a direct implication of the neighborhood-level neglect documented in the previous section, this relies on the assumption that neglect doesn’t change under these selection environments. A priori, officers might be better at adjusting for selection when there is clear bias across groups, or when observability is affected by their predictions. We directly test this hypothesis, and find that officers are not better at adjusting for selection when it is endogenous or shows group bias. As a result, selection bias across groups translates into statistical discrimination, and endogenous selection makes this bias persistent. Even in the absence of stereotypes, incentives, or taste, differences in observability generate disparities in crime predictions.

Selection Bias Across Groups

Crime observability differ systematically across locations and groups. In the United States, Black and Hispanic individuals are overrepresented in crime data because they are more likely to be patrolled, stopped, reported in a field interview, and arrested (Gonçalves, Mello, & Weisburst, 2025; National Academies of Sciences, Engineering, and Medicine, 2023). In Colombia, crime reporting rates vary across areas and income groups (DANE, 2023). These uneven selection patterns imply there is selection bias across groups: some groups are more likely to enter administrative crime data than others. If officers neglect selection under this conditions, selection bias will generate systematic bias in predictions across groups, this is, statistical discrimination.

We directly test this implication by introducing a new treatment with selection bias across cities. In the baseline *Uniform Selection* environment, selection is identical across cities: reporting rates in both cities are drawn from the same distribution, so the expected reporting-rate gap is zero. In contrast, in

the *Selection Bias Across Groups* treatment, reporting rates for the two cities are drawn from different distributions, generating an expected reporting-rate gap of 30 percentage points. Importantly, underlying crime distributions remain identical across cities. Thus, in this treatment cities systematically differ in the number of crimes observed (reported), but this difference is entirely driven by observability. Under this design, systematic prediction bias across cities can be solely attributed to selection neglect. We randomize officers either into the Baseline or the Selection Bias Across Groups condition for all twelve decisions of Part II.

A priori, it is possible that police officers are better at recognizing these patterns of selection bias than random variation at the neighborhood level, as they are plausibly more salient. However, we rule out this possibility: selection neglect remains stable under this environment. Table A.8 shows that officers continue to neglect selection when it shows bias across groups, with magnitudes comparable to those observed under uniform selection. That neglect remains stable despite selection being plausibly more salient is consistent with the results of Section V and reinforces the interpretation that neglect reflects a stable mental representation of the inference problem rather than transient inattention.

Because neglect remains stable, selection bias across groups causes biased beliefs. Because crime is identical across cities, a rational decision maker would show no prediction bias across them. In contrast, a neglecting decision maker would predict higher crime where it is more observable. Panel (a) of Figure 4 tests this prediction by plotting prediction bias as the difference in prediction probability between the city with higher average observability and the one with lower observability¹⁰. A prediction bias of zero indicates that officers are equally likely to predict crime between cities with different observability, while positive values indicate that officers take observable crime for true crime. Under uniform selection, with no difference in observability across cities, we find no prediction bias across groups ($p = 0.852$). In contrast, officers in the Selection Bias Across Groups condition show systematic prediction bias: they are 20 percentage points ($p = 0.006$) more likely to predict higher crime in the group with higher observability, despite identical true crime levels.

Defining prediction bias more broadly offers a similar picture. We first calculate the officer-level optimal benchmark: how often their optimal prediction would be city C. Note that, although on expectation both cities have equal crime, because of randomization the benchmark is not 50% for all officers. We then calculate prediction bias as the absolute deviation from this optimal benchmark, in percentage points. For instance, an officer that predicts higher crime in city C in 33% of decisions, when their optimal benchmark was 50%, will have a prediction bias of 17 percentage points. Figure A.1 plots prediction bias for both treatments. When selection is uniform across groups, officers show a bias of 17.8 percentage points. Selection bias across groups increases prediction bias by over 9 percentage points ($p = 0.006$). Again, despite crime levels not changing across treatments, selection bias across groups amplifies disparities in beliefs.

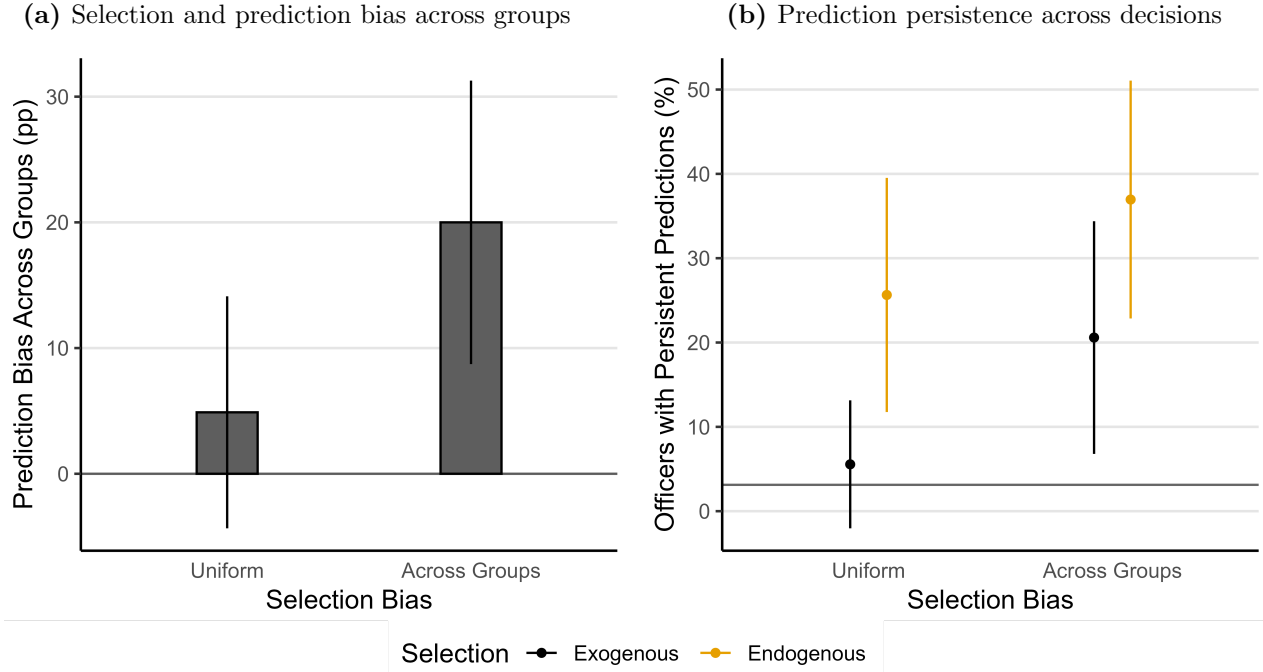
These biased beliefs are persistent. Panel (b) of Figure 4 shows the share of officers who persistently predict one city to have higher crime across the last six decisions of Part II. When selection is uniform across groups, only 5.5% of officers make persistent predictions, a fraction broadly consistent with opti-

¹⁰At baseline, both cities have the same reporting rate in expectation. However, because of randomization, one city always has a slightly higher reporting rate on average. We take this city as the city with higher average observability at baseline.

mality¹¹. In contrast, 20.6% of officers in the Selection Bias Across Groups condition predict the same city to have more crime in all six decisions. Table A.9 confirms that these differences are statistically significant.

Taken together, these findings show that when observability is biased across groups, as is common in crime data, selection neglect translates that selection bias into persistent disparities in predicted crime. Even in the absence of group-specific preferences, incentives, or differences in true crime, selection bias is sufficient to generate statistical discrimination in beliefs.

Figure 4: Prediction bias and persistence across selection patterns



Notes: Panel (a) shows the prediction bias across groups: the difference in prediction probability between the city with higher average reporting rate and the city with lower reporting, in percentage points. For instance, a prediction bias of 20 percentage points indicates that the average participant is 20 points more likely to predict more crime in the city with higher reporting rates than in the alternative. The left column shows officers in the *Uniform Selection* treatment, where reporting rates for both cities are equal in expectation. The right column shows officers in the *Selection Bias Across Groups* treatment, where reporting rates are 30 percentage points higher in expectation for one city. The figure only includes the first six predictions of Part II. Error bars show the 95% confidence interval of the difference in means, with standard errors clustered by officer. The horizontal black line at zero shows the optimal benchmark: because both cities have the same number of crimes in expectation, optimal predictions should be unbiased across cities. Panel (b) shows the share of officers who make *persistent predictions*, this is, consistently predict the same city to have higher crime across the last six decisions of Part II. This share is disaggregated across selection environments. The left column shows officers in the *Uniform Selection Bias* treatment, while the right column shows officers in the *Selection Bias Across Groups* treatment. Colors represent the endogeneity treatments. In black, we show officers in the *Exogenous Selection* group, where crime predictions don't affect reporting rates, and in orange the *Endogenous Selection* treatment, where predicting higher crime in one city increases reporting rates by 15 percentage points in that same city for the next decision. Error bars show the 95% confidence interval of the means, with standard errors clustered by officer. The diamonds show the optimal benchmark: because crimes are equally distributed across cities, the chance that it's optimal to guess the same city in 6 consecutive decisions is $2 \times 0.5^6 = 3.1\%$.

¹¹Because crimes are equally distributed across cities, the chance that it's optimal to guess the same city in 6 consecutive decisions is $2 \times 0.5^6 = 3.1\%$.

Endogenous Selection

We next examine selection neglect when observability depends on past predictions. In many policing settings, crime predictions influence patrols, stops, and searches, which affect which crimes are observed (Brayne, 2020; Chen et al., 2023). When predictions influence what becomes observable, data selection is endogenous. If officers neglect endogenous selection, prediction errors can feed back into the data-generating process and transform into self-reinforcing cycles of bias. However, this amplification depends on selection neglect remaining stable when selection becomes endogenous rather than exogenous.

We experimentally test this implication by introducing an additional treatment in which predictions affect future observability. In the *Endogenous Selection* treatment, predicting higher crime in a given city increases its reporting rate in the subsequent decision by 15 percentage points. In contrast, in the baseline *Exogenous Selection* condition reporting rates are drawn independently of past predictions. Crucially, in both conditions the underlying crime distributions are identical across cities. This design isolates the effect of endogenous observability while holding true crime constant.

We first verify that selection neglect remains stable when selection is endogenous. Table A.8 confirms that officers continue to treat observed crime as representative of true crime, even when their own predictions influence observability. Because neglect remains stable, endogenous selection generates persistent prediction cycles. First, endogenous selection increases short-run persistence in predictions. Because crime is identical across cities, there is a 50% chance that the same city has higher crime in two consecutive predictions. Thus, a rational and a random decision maker would be equally likely to repeat or switch cities no matter whether selection is exogenous or endogenous. In contrast, a neglecting decision maker would be more likely to repeat under endogenous selection, as each prediction increases observability in that city for the next period. Table A.9 regresses a dummy of whether each prediction is repeated on the Endogenous Selection Treatment. At baseline, officers are equally likely to repeat or switch. Under endogenous selection, officers become 14.6 percentage points ($p = 0.027$) more likely to consecutively predict crime in the same group. This increase reflects the feedback implied by neglect: once a city is predicted to have higher crime, higher subsequent observability reinforces that belief. This short-run persistence compounds into longer-run disparities in beliefs. Figure 4 shows that under endogenous selection, officers are 21 percentage points ($p = 0.065$) more likely to predict the same city across all six decisions, even though selection is a priori uniform across groups and underlying crime is identical across cities.

In many policing contexts, selection bias across groups and endogeneity in crime data coexist. In these cases, endogenous selection might reinforce the statistical discrimination arising from selection bias and generate persistent cycles of bias. To explore the combined effects of these features we focus now on officers who were simultaneously randomized to the Selection Bias Across Groups and Endogenous Selection conditions for the last six decisions. Column (2) in Table A.9 shows that under both selection features, officers become 17.6 percentage points more likely to patrol the city with higher observability than at baseline ($p = 0.054$), leading to a 28 percentage point prediction bias across groups on average. Because of endogenous selection, this bias grows over time. In the last decision before endogenous selection is introduced, there is roughly a 8 percentage point bias in favor of the city with higher observability. After six endogenous predictions, this bias grows to 40 percentage points. Finally, selection-induced belief bias becomes more persistent when selection is endogenous and biased across groups. Columns (3) and (4) in Table A.9 regress the likelihood that an officer always predicts higher crime in the same city on

the selection treatments, and find these features increase that likelihood by around 30 percentage points ($p < 0.001$).

Taken together, these results show that selection neglect systematically converts selection bias into prediction bias. When observability is biased across groups, neglect translates those differences into persistent disparities in predicted crime. When predictions influence future observability, neglect generates self-reinforcing cycles that amplify and sustain those disparities over time. Crucially, these patterns arise even though underlying crime distributions are identical and group identities are abstract, ruling out preferences, stereotypes, or group-specific priors as explanations. Selection neglect therefore provides a microfoundation for statistical discrimination: biased and endogenous selection is sufficient to generate persistent belief disparities in the absence of animus.

V Mechanisms

A Framework: Two Margins of Selection Neglect

We now delineate a simple conceptual framework to build intuition about the possible mechanisms of selection neglect. Consider a decision maker who uses selected data r and a measure of selection s to infer another value c , and let their inference rule be $f(\cdot)$. In our experimental setting, r corresponds to the number of reports, s to the reporting rate, c is the number of crimes, and the optimal inference rule is $f^*(r, s) = r/s$. Selection neglect can be decomposed in two distinct cognitive failures. First, a decision maker can fail to incorporate selection into their inference by implementing some rule $f(r)$. Second, conditional on incorporating selection, they might not apply the correct inference rule: $f(r, s) \neq f^*(r, s)$. These two distinct margins are separable in our design.

The first margins for neglect corresponds to selection failing to “come to mind” when making inference based on selected data. While selection neglect might be attributed to lack of information about the selection process, a decision maker who is not aware of selection will not use this information even when available, as their inference problem doesn’t demand it as an input. We directly measure the prevalence of failures at this margin by giving officers the choice between looking at reporting rates or some other irrelevant features of the problem.

There are two non-exclusive explanations for ignoring selection altogether: bottom-up and top-down inattention. When representing the inference problem, salience might drive bottom-up attention to other features of the data at the expense of selection (Bordalo et al., 2023). In other words, the decision maker represents the problem and chooses the inference rule $f(\cdot)$ depending on the relative salience of different features. However, while salience might affect the representation of novel problems, it is possible that officers already hold a established representation coming from their prior experience with the inference problem. If this representation and its inference rule $f(r)$ don’t use selection as an input, the decision maker will be inattentive to information about selection s (Maćkowiak et al., 2023; Schwartzstein, 2014). We test the role of both sources of inattention in our experiment. We exogenously vary the salience of observability: while in Part I reporting rates are embedded among other irrelevant data, in Part II officers only see the number of reports and reporting rates, making the latter more salient and difficult to miss. If salience is driving neglect, we expect higher salience to reduce this cognitive failure. Finally, we use previous experience predicting crime from selected reports data to offer suggestive evidence of whether mental models explain inattention.

The second margin of neglect arises when selection is *imprecisely* incorporated into the inference problem. For instance, when inferring quality from product reviews, a consumer might know to discount the average rating of a product with few reviews, but not by how much. Thus, the decision maker might take selection into account but use the wrong inference rule $f(r, s) \neq f^*(r, s)$. In our experiment, an officer might realize that reporting rates matter to infer underlying crime, but might not use this information precisely. We explore this margin by measuring neglect among the sample who chose to view reporting rates over other irrelevant features, this is, the prevalence of wrong inference rules conditional on these using selection as an input.

A plausible explanation for incorrectly adjusting for selection is that individuals do not know the precise data generating process. In the previous example, the consumer doesn’t perfectly know who

self-selects into writing a product review. While we make the data generating process transparent to all participants and test whether they understand it, officers might bring to the lab the mental representation they have developed in the field. In other words, officers who are more aware of selection bias in the crime data they routinely use might be better at adjusting for selection in our experiment. We directly test this hypothesis by correlating neglect to officers’ reported awareness of selection bias in crime data in the field. Alternatively, the decision maker might know the correct inference rule but fail to implement it due to computation costs. We show suggestive evidence of this mechanism by looking at response times and on-screen calculator use.

Finally, we use algorithmic recommendations to further separate mental models, salience, and computation costs. A recommendation from an algorithm that implements the correct inference rule $f^*(r, s)$ should eliminate neglect if this is a computational failure, as the algorithm already makes the calculation for the officer. However, the recommendation might be overridden if the officer sticks to their wrong representation, making the algorithm ineffective.

B Mental Models Explain Failures at Both Margins

We begin by examining the first margin of neglect: whether selection enters the officers’ representation of the problem at all. In the tenth decision of Part I, officers choose which variable to view alongside reported crime. Choosing to view the reporting rate indicates that selection enters their prediction. Figure A.5 finds that 36% of officers choose not to view the reporting rate, even though the data-generating process is completely transparent. To test the robustness of this result, Table A.6 regresses the choice to view selection on instruction comprehension, math skills, professional experience, and awareness of selection bias in crime data. We find that choosing to ignore selection is not explained by confusion about the task or lack of basic numeracy. Instead, it varies systematically with officers’ previous assignment. Officers whose most recent assignments involved surveillance and patrol assignment (tasks that rely heavily on predicting crime from reported crime data) are 20 percentage points ($p = 0.035$) more likely to view reporting rates than administrative officers.

Because officers with more experience making crime predictions from selected data have plausibly developed better inference models, these results suggest the role of top-down inattention in ignoring selection. We next explore this role in further detail by regressing officers’ predictions before the choice to view selection on the differences between all available variables, this is, the gap between neighborhoods in number of crime reports, reporting rates, and demographic decoys. We interact each of these with the later choice to view selection. A significant relationship between one of these variables and predictions indicate that officers use this information to guide their predictions. Table A.4 confirms that the 36% of the sample who choose not to view reporting rates didn’t use this variable to make crime predictions, as predictions don’t change with the gap in reporting rates ($p = 0.325$). Instead, these officers rely on reported crime data, without adjusting for selection, and on the uninformative demographic decoys. This doesn’t only confirm that information demand corresponds to actual information use, but suggests that officers ignore selection because they don’t see a need for this information. Together with the correlation between field experience and information choice, these findings point to top-down inattention playing an important role in selection neglect: officers whose mental representations don’t include as an input will simply ignore it.

An alternative explanation for failures at the first margin is salience. In the field, officers might ignore reporting rates because these are relatively opaque, and might focus instead on more accessible information. In our experiment, officers might ignore reporting rates when they’re embedded among demographic decoys. To test this explanation, we compare crime predictions while exogenously varying the salience of reporting rates. In Part I, reporting rates and crime reports are shown alongside multiple uninformative variables; in Part II, these decoys are removed, making selection visually unavoidable. We first focus on the officers who previously chose not to view selection and find that increasing its salience doesn’t make them better at adjusting for selection. Column (6) of Table A.4 shows that these officers don’t adjust their predictions by the gap in reporting rates even when selection is made more salient in Part II ($p = 0.434$). As a result, increasing salience fails to debias crime predictions. Table A.1 documents that selection causes an excess in prediction errors of 32 percentage points among those officers in Part II, up from an excess of 28 percentage points when selection is not salient in Part I. Salience has no effect either among officers who already incorporated selection into their inference process. In aggregate, selection neglect is no different across levels of selection salience.

Overall, over a third of our sample completely ignores selection, and this decision seems driven by wrong inference rules $f(r)$ rather than by salience. Selection fails to come to mind not because it is unseen, but because it is not represented as a relevant feature of the inference problem. As such, interventions that aim to bring selection to the mind of officers when making crime predictions are likely to be ineffective unless they also target officers’ mental models.

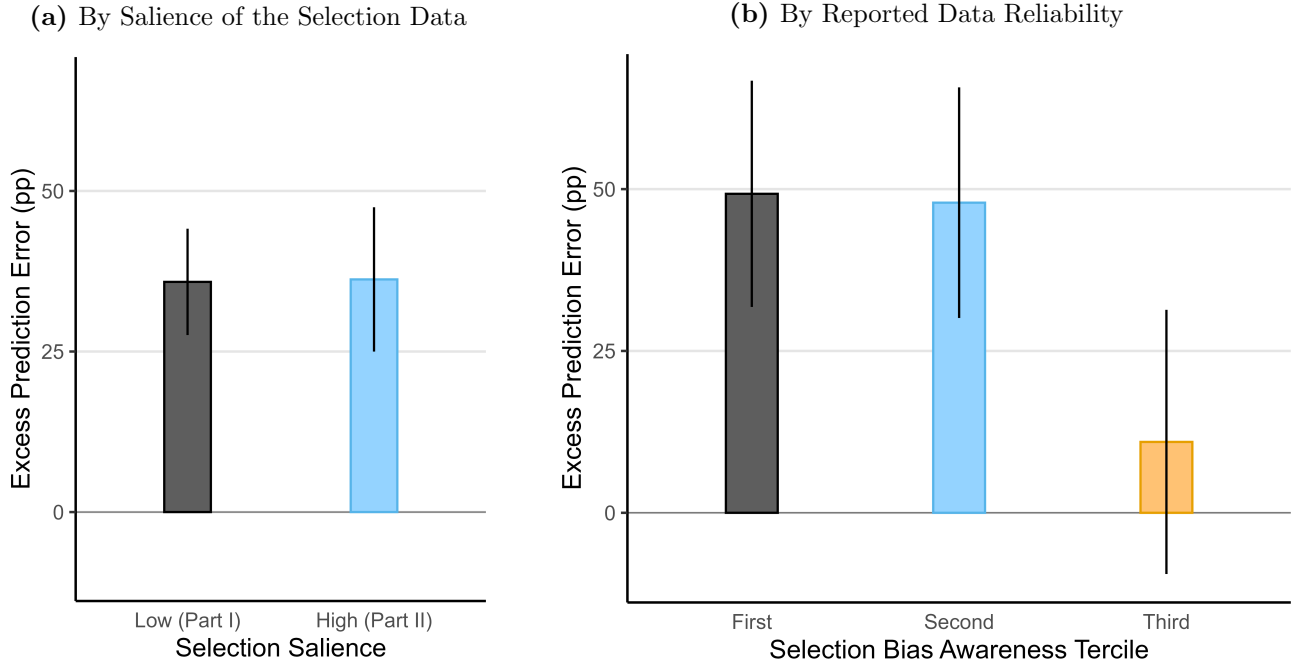
While about one third of officers fail to take selection into account at all, this margin cannot fully explain the neglect we document. Table A.1 documents that, even among the two thirds who explicitly recognize that reporting rates matter, selection still increases prediction errors. In other words, neglect persists even when selection does come to mind. What drives failures at the second margin? There are two broad possibilities. First, officers may hold an incorrect understanding of how reporting shapes observed crime and how to properly adjust for it, that is, they rely on the wrong inference rule $f(r, s)$. Second, officers may understand the correct adjustment $f^*(r, s)$ but fail to apply it consistently because correcting for selection is cognitively demanding. These explanations generate different predictions. If neglect reflects flawed mental models, it should vary systematically with officers’ understanding of field data. If it reflects cognitive frictions, it should primarily be driven by effort and computational constraints. We now test these competing mechanisms.

If selection neglect reflects inaccurate mental models, then officers who better understand how crime data are generated in the field should be less prone to neglect in the lab. We test this hypothesis by examining whether neglect varies with officers’ awareness of selection bias in the crime data they routinely work with. At the end of the experiment, officers report their agreement, on a scale from 1 to 5, with five statements about selection bias in reported crime data. These statements ask whether crime reports accurately reflect spatial and temporal patterns, whether citizens know how and where to report crimes, and whether reported data provide a biased picture of underlying crime. We aggregate these responses into an index that we denote *field selection awareness*, where higher values reflect greater recognition that reported crime data are selected through reporting behavior ¹².

Consistent with the mental model explanation, officers who are more aware of selection bias in field

¹²We follow Kling et al. (2007) and aggregate z-scores of each component of the index.

Figure 5: Heterogeneity in Selection Neglect



Notes: Panel (a) shows the accuracy loss due to selection —the estimated coefficient of selection-sensitivity on accuracy, disaggregated by selection salience. The left bar plots the effect for Part I, with low selection salience, and the right bar plots it for Part II, with higher selection salience. Error bars show the 95% confidence interval of the coefficient. Panel (b) shows the accuracy loss due to selection, disaggregated by reported data reliability. This panel only uses prediction from Part II. We plot the coefficient of selection-sensitivity estimated separately by terciles of reported data reliability. Error bars show the 95% confidence interval of the coefficient. Diamonds show the optimal benchmark in both panels.

data are better at adjusting for selection in our experiment. We divide the sample into terciles based on field selection awareness and estimate the excess prediction error caused by selection separately for each group. Panel (b) of Figure 5 shows that among the third of officers with the highest awareness, selection increases prediction error by only 10 percentage points ($p = 0.298$). In contrast, excess prediction error reaches nearly 49 percentage points among the third of the sample with the lowest awareness. These differences suggest that correctly adjusting for selection requires more than simply attending to reporting rates: it requires an accurate understanding of how reporting shapes the observability of crime.

An alternative explanation is that neglect at the intensive margin reflects cognitive frictions: officers may represent the problem correctly, but fail to implement the rule $f^*(r, s)$ because it is more costly. We find little evidence to support this hypothesis. First, an algorithm that eliminates calculation costs doesn't improve crime predictions, as explained in more detail in the next subsection. Second, the correct calculation only involves a simple division, and participants have a calculator on screen. However, only 30% of the sample ever use it in Part II, and this choice is positively correlated with field selection awareness ($p = 0.024$). Thus, although Table A.1 finds no neglect among participants who used the calculator, we can't rule out this better performance is uniquely driven by better mental models. Overall, the evidence suggests that officers who represent the problem correctly use the calculator to reduce frictions, but these frictions definitely can't account for the excess prediction error caused by selection.

Overall, these findings indicate that mental representations rather than computation frictions or salience explain selection neglect among police officers. While we find that mental models explain failures at both margins of neglect, we find that boosting salience or reducing calculation costs don't affect neglect. A natural implication is that increasing awareness of selection bias in crime data could reduce neglect. However, the evidence from field selection awareness cautions against such a simple interpretation. We regress crime predictions in Part I on the differences between available variables (crime reports, reporting rates, and demographic information), and estimate this regression separately for each tercile of field selection awareness. Figure A.2 finds that officers who are more aware of selection bias rely more on reporting rates, but also substitute reported crime data for socioeconomic variables that have no predictive power in our setting. Improving awareness may therefore correct one distortion while introducing another.

C Selection Neglect and Advice Override

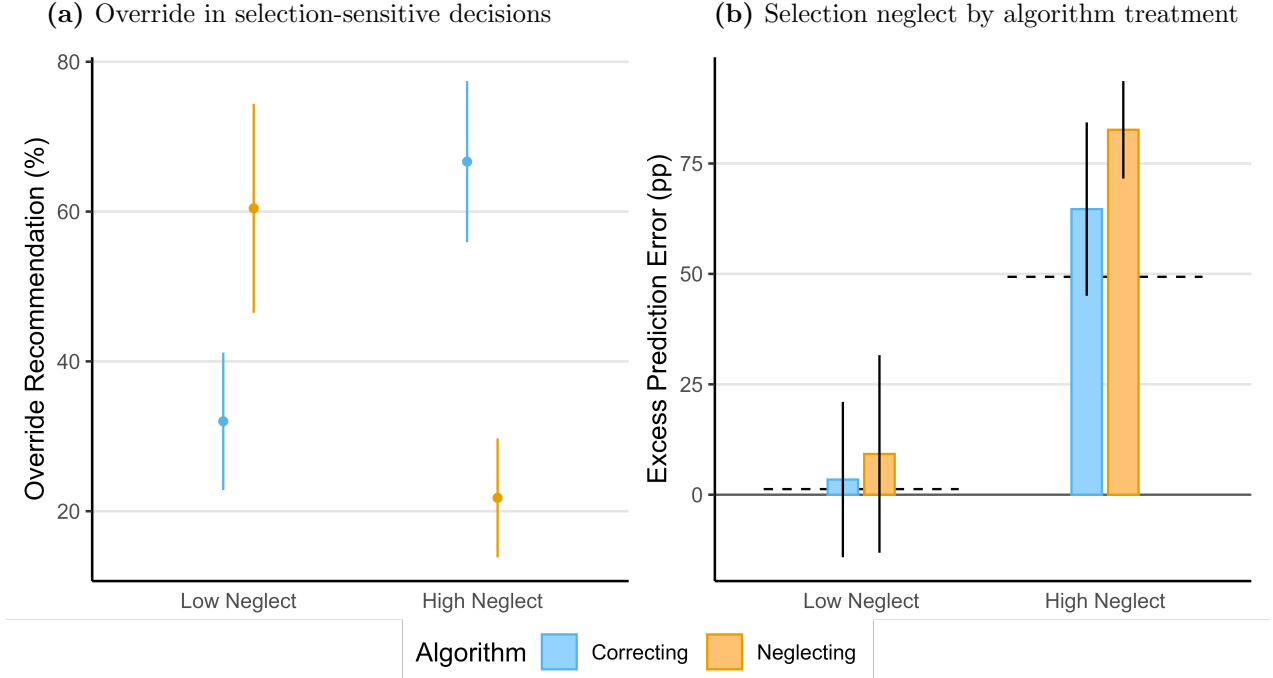
We next introduce algorithmic recommendations as a test of the mechanism underlying selection neglect. For the second half of Part II, officers are randomly assigned to one of three conditions: a control group with no recommendation, a correcting algorithm that adjusts reported crime for reporting rates ($f^*(r, s)$), and a neglecting algorithm that relies on raw reports ($f(r)$). Officers are informed whether their algorithm accounts for selection and retain full discretion to follow or override its recommendations. Because officers retain discretion, adherence and override patterns reveal whether algorithms substitute for inference or are filtered through existing rules.

Our conceptual framework considers three overarching explanations for selection neglect: salience, inaccurate mental models, and computational frictions. Under a salience account, officers fail to adjust for selection because it does not come to mind. In this case, algorithmic recommendations should be followed regardless of whether they correct for selection, as they're the more salient feature of the decision. Thus, the correcting algorithm should reduce neglect. Under a computational account, officers know the correct rule $f^*(r, s)$ but fail to apply it; the correcting algorithm should therefore reduce neglect by eliminating the need to perform the calculation, particularly among officers with high pre-algorithm neglect. The neglecting algorithm should thus have no effect, because officers would know it's implementing the wrong rule. In contrast, under a mental-model account, officers follow the inference rule they believe to be correct under their representation of the prediction problem. High-neglect officers, whose pre-algorithm behavior reveals reliance on a distorted rule, should follow the neglecting algorithm, which validates their rule, and override the correcting algorithm, which contradicts it. Low-neglect officers should display the opposite pattern. These mechanisms thus generate distinct predictions for both adherence and changes in neglect under the two algorithms.

We begin by dividing the sample based on pre-algorithm behavior. Using the first six decisions in Part II, before algorithms are introduced, we classify officers into two groups: those with above the median prediction error in selection-sensitive decisions (high neglect), and those below the median (low neglect). This classification broadly captures whether the inference rule officers were using prior to receiving any algorithmic guidance were correct. We then examine how each group responds to the correcting and neglecting algorithms.

We first analyze whether officers follow or override recommendations. Panel (a) of Figure 6 reports the fraction of algorithmic recommendations that officers choose to override. Among officers with low

Figure 6: Selection neglect and adherence to algorithm recommendations



Notes: Panel (a) shows the share of selection-sensitive decisions in which officers override the algorithm recommendation, dividing the sample in half according to their level of neglect in the first six decisions of Part II. The left column shows officers with Low Neglect —below-median prediction error ($< 66\%$) in selection-sensitive decisions, while the right column shows officers with High Neglect —above-median prediction error. Error bars show the 95% confidence interval of the mean, with standard errors clustered at the officer level. Panel (b) shows the excess prediction error due to selection —the estimated coefficient of selection-sensitivity on prediction error, disaggregated by algorithm intervention and level of neglect. The dashed lines represent the excess prediction error of officers in each group that were randomized to the control algorithm (no algorithm advice). Error bars show the 95% confidence interval of the coefficient, with standard errors clustered at the officer level. Both panels only use the last six decisions of Part II. Colors show the algorithm treatments: Control in black dashed lines, Correcting in blue, and Neglecting in orange.

pre-algorithm neglect, the correcting algorithm is followed substantially more often than the neglecting one: they override the neglecting algorithm in roughly 60% of cases but override the correcting algorithm in only about 30%. In contrast, officers with high pre-algorithm neglect display the opposite pattern. They override the correcting algorithm in roughly two thirds of decisions, yet override the neglecting algorithm in only about 20%. Thus, officers selectively adhere to algorithmic advice when it aligns with their prior inference rule and reject it when it contradicts it. Finally, Figure A.7 shows that higher field selection awareness is associated with more override of the neglecting algorithm and more adherence to the correcting one. The link between field selection awareness and override patterns reinforces the mental model interpretation of these results.

We next examine how these adherence patterns translate into neglect-induced prediction errors. Panel (b) of Figure 6 reports excess prediction error by group and algorithm. Among officers with low pre-algorithm neglect, excess errors remain close to zero across all conditions. Among officers with high pre-algorithm neglect, however, the correcting algorithm does not reduce excess errors relative to the control group —it slightly increases them, ($p = 0.264$), while the neglecting algorithm increases them by over 30 percentage points ($p = 0.003$). Because officers only follow recommendations aligned with their

preconceived inference rule, algorithmic advice fails to correct neglect. Instead, it might amplify neglect by validating the wrong rule.

Taken together, these results reject both a salience and a computational explanation. If neglect were driven by salience, high-neglect officers would follow both algorithms. If it were driven by computational frictions, the correcting algorithm should reduce errors for this group. Instead, algorithmic advice is filtered through existing mental representations. Information provision that does not change the underlying representation of how crime data are generated is therefore unlikely to eliminate selection neglect, and may even reinforce it.

VI Discussion

This paper studies how police officers form crime beliefs from data that are not representative of underlying crime. We show that officers systematically treat observed crime as if it were representative, failing to account for how crime becomes observable. This selection neglect leads to large and economically meaningful errors in crime predictions. Importantly, these errors translate selection bias across groups into biased and persistent beliefs about crime. In this way, a simple cognitive failure in belief formation can generate systematic disparities in crime predictions and data-driven policing.

Over the past decades, policing has become increasingly data-driven, in part with the aim of reducing disparities associated with stereotypes or animus (Brayne, 2020). Yet substantial racial and socioeconomic disparities persist (National Academies of Sciences, Engineering, and Medicine, 2023). Our results identify a distinct channel through which such disparities can arise. When crime data reflect systematic differences in observability across groups, and officers fail to adjust for this selection, differences in data visibility translate into differences in predicted crime even when underlying crime is identical. This mechanism generates inaccurate statistical discrimination: disparities in beliefs driven by a cognitive bias rather than by preferences or correct statistical inference. While our results do not rule out a role for taste-based discrimination, they show that disparities can emerge solely from how decision-makers process selected data. This approach aligns with a growing literature focusing on the cognitive aspects of policing. For example, Ferrazares (2025) show that cognitive depletion due to consecutive shifts leads to more discretionary decisions and more use of force, and Dube et al. (2024) and Owens et al. (2018) find that training officers to deliberate before making decisions limits those same outcomes.

By identifying selection neglect at the stage of belief formation, our results provide a microfoundation for statistical discrimination in policing. Policing occupies a central position in the criminal justice data pipeline: beliefs about where crime is more likely guide patrol allocation and enforcement, which in turn determine which crimes are observed and recorded. These data then inform the beliefs and decisions of downstream actors, such as prosecutors and judges. When beliefs are formed from selected data and selection is neglected, distortions arise at the source and can propagate through subsequent stages of inference and decision-making throughout the criminal justice system.

A growing literature studies the role of beliefs among decision makers in the criminal justice system (Arnold et al., 2018; Bhuller & Sigstad, 2024; Feigenberg & Miller, 2025). While beliefs lie upstream of enforcement decisions, focusing on how officers form beliefs about crime provides several advantages. The most important one is the ability to disentangle the different mechanisms behind selection neglect. Our evidence suggests that selection neglect is primarily a failure of representation rather than of attention or computation. Officers either fail to treat selection as a relevant feature of the inference problem or apply an incorrect adjustment rule even when selection is salient and fully observable. Consistent with this interpretation, increasing the salience of reporting rates or providing algorithmic recommendations that implement the correct adjustment does not improve predictions, as officers frequently override advice that conflicts with their prior beliefs. These findings, which resonate with previous work on algorithmic override among judges (Angelova et al., 2025), have direct implications for policy. Interventions that provide more information or reduce computational burden are unlikely to correct bias if they do not change how decision-makers model the inference process. Instead, approaches that reshape mental models, such as training that explicitly links observability to crime data, may be more effective. At the same time, our results

suggest caution: officers who are more aware of selection bias also express lower trust in reported crime data and may substitute toward other, potentially uninformative, signals. Effective interventions must therefore correct misinterpretation of data without inducing generalized distrust in official information.

Our cognitive approach to policing builds on extensive evidence of selection neglect among standard experimental subjects (Ali et al., 2021; Araujo et al., 2021; Enke, 2020; Esponda & Vespa, 2018; Farina & Herman, 2025; Jin et al., 2021). Studying experienced practitioners allows us to extend this work in two important ways. First, we show that selection neglect is pervasive even in among professionals who routinely work with selected data in a high-stakes setting. Second, running a framed field experiment with practitioners allow us to leverage heterogeneity in experience and domain-specific beliefs, which would not be possible with experimental subjects. Our findings contribute to this literature by bringing the lab to field practitioners and confirming that selection neglect operates among them, consistent with prior work from the financial sector (Koehler & Mercer, 2009; Malmendier & Shanthikumar, 2007) and the formation of political worldviews (Cruces et al., 2013; DellaVigna & Kaplan, 2007). Additionally, this paper tests and confirms, with the relevant population, the theoretical predictions of Che et al. (2024) and Hübert and Little (2023), who suggest the mechanism of neglect as a driver of policing disparities.

The dynamics we document are not specific to policing, but arise more generally in environments where data are selectively observed and endogenously generated. In any domain where search of new information is prediction-based (Caplin et al., 2011; Malladi et al., 2025; Reutskaja et al., 2011), current predictions influence which information become observable in the future, creating feedback between beliefs and data. When inference errors are stable, such environments give rise to cycles of bias in which distorted beliefs shape data generation and selected data reinforce those beliefs. This logic is so general that it applies not only to human decision-makers, but also to predictive algorithms. Evidence from predictive policing and criminal risk assessment tools shows that algorithms trained on selected crime data can reproduce and amplify existing biases, due to selection neglect (Arnold et al., 2025; Lum & Isaac, 2016). A fruitful avenue for future research lies in extending the analysis of selection neglect to other contexts where data suffers from endogenous selection bias, like healthcare, hiring, or finance.

References

- Aggarwal, P., Brandon, A., Goldszmidt, A., Holz, J., List, J. A., Muir, I., Sun, G., & Yu, T. (2025). High-frequency location data show that race affects citations and fines for speeding. *Science*, 387(6741), 1397–1401.
- Ali, S. N., Mihm, M., Siga, L., & Tergiman, C. (2021). Adverse and Advantageous Selection in the Laboratory. *American Economic Review*, 111(7), 2152–2178.
- Ang, D., Bencsik, P., Bruhn, J., & Derenoncourt, E. (2025). Community Engagement with Law Enforcement after High-Profile Acts of Police Violence. *American Economic Review: Insights*, 7(1), 124–142.
- Angelova, V., Dobbie, W., & Yang, C. S. (2025). Algorithmic Recommendations and Human Discretion. *The Review of Economic Studies*, rdaf084.
- Araujo, F. A., Wang, S. W., & Wilson, A. J. (2021). The Times They Are A-Changing: Experimenting with Dynamic Adverse Selection. *American Economic Journal: Microeconomics*, 13(4), 1–22.
- Arnold, D., Dobbie, W., & Hull, P. (2025). Building Nondiscriminatory Algorithms in Selected Data. *American Economic Review: Insights*, 7(2), 231–249.
- Arnold, D., Dobbie, W., & Yang, C. S. (2018). Racial Bias in Bail Decisions. *The Quarterly Journal of Economics*, 133(4), 1885–1932.
- Ba, B., Knox, D., Mummolo, J., & Rivera, R. (2021). The role of officer race and gender in police-civilian interactions in Chicago. *Science*, 371(6530), 696–702.
- Barrios, J. M., Lancieri, F., Levy, J., Singh, S., Valletti, T., & Zingales, L. (2025, April). *The Conflict-of-Interest Discount in the Marketplace of Ideas* (tech. rep. No. w33645). National Bureau of Economic Research.
- Barron, K., Huck, S., & Jehiel, P. (2024). Everyday Econometricians: Selection Neglect and Overoptimism When Learning from Others. *American Economic Journal: Microeconomics*, 16(3), 162–198.
- Bhuller, M., & Sigstad, H. (2024). Feedback and Learning: The Causal Effects of Reversals on Judicial Decision-Making. *The Review of Economic Studies*, rdae073.
- Bordalo, P., Coffman, K., Gennaioli, N., & Shleifer, A. (2016). Stereotypes. *The Quarterly Journal of Economics*, 131(4), 1753–1794.
- Bordalo, P., Conlon, J., Gennaioli, N., Kwon, S., & Shleifer, A. (2026). How People Use Statistics. *The Review of Economic Studies*, 93(1), 250–285.
- Bordalo, P., Conlon, J. J., Gennaioli, N., Kwon, S. Y., & Shleifer, A. (2023). How People Use Statistics. *NBER Working Paper*.
- Brayne, S. (2020, November). *Predict and Surveil: Data, Discretion, and the Future of Policing*. Oxford University Press.
- Brundage, M., Little, A. T., & You, S. (2024). Selection Neglect and Political Beliefs. *Annual Review of Political Science*, 27(Volume 27, 2024), 63–85.
- Caplin, A., Dean, M., & Martin, D. (2011). Search and Satisficing. *American Economic Review*, 101(7), 2899–2922.
- CEJ. (2025). *Informe de estadísticas del SPOA 2025* (tech. rep.).
- Che, Y.-K., Kim, J., & Mierendorff, K. (2024, September). Predictive Enforcement.
- Chen, M. K., Christensen, K. L., John, E., Owens, E., & Zhuo, Y. (2023). Smartphone Data Reveal Neighborhood-Level Racial Disparities in Police Presence. *The Review of Economics and Statistics*, 1–29.
- Cruces, G., Perez-Truglia, R., & Tetaz, M. (2013). Biased perceptions of income distribution and preferences for redistribution: Evidence from a survey experiment. *Journal of Public Economics*, 98, 100–112.
- DANE. (2023). *Encuesta de convivencia y seguridad ciudadana (ECSC)* (tech. rep.).
- DellaVigna, S., & Kaplan, E. (2007). The Fox News Effect: Media Bias and Voting. *The Quarterly Journal of Economics*, 122(3), 1187–1234.
- Dube, O., MacArthur, S. J., & Shah, A. K. (2024). A Cognitive View of Policing. *The Quarterly Journal of Economics*.
- Enke, B. (2020). What You See Is All There Is. *The Quarterly Journal of Economics*, 135(3), 1363–1398.
- Esponda, I., Oprea, R., & Yuksel, S. (2023). Seeing What is Representative. *The Quarterly Journal of Economics*, 138(4), 2607–2657.

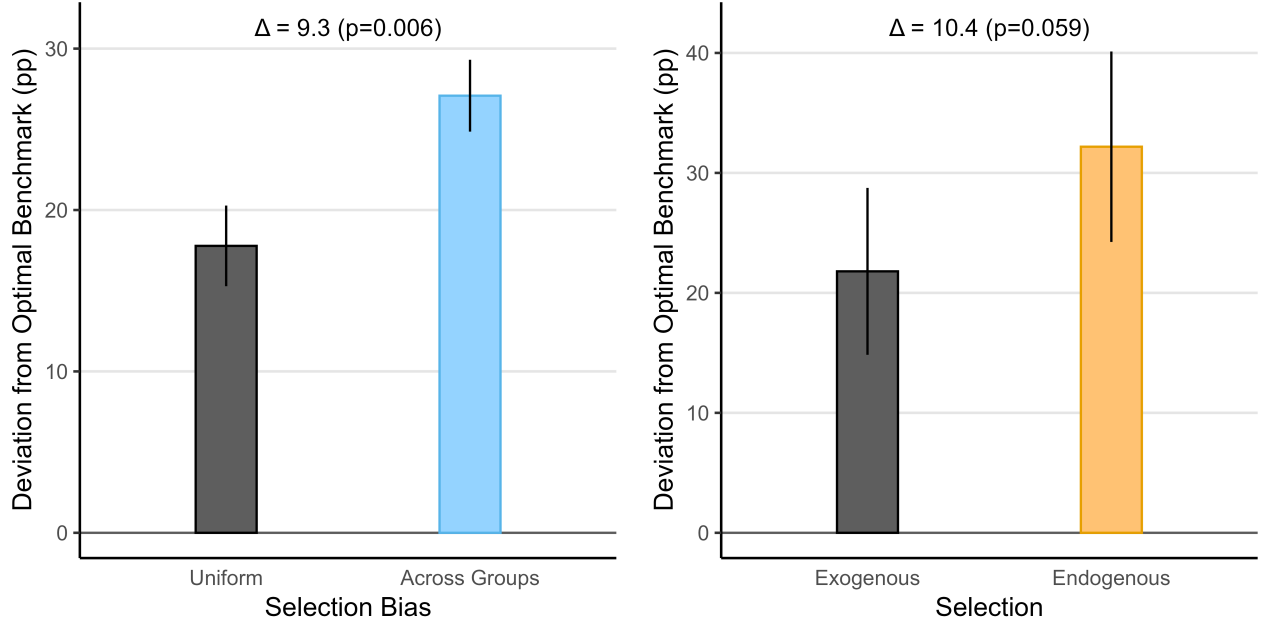
- Esponda, I., & Vespa, E. (2018). Endogenous sample selection: A laboratory study. *Quantitative Economics*, 9(1), 183–216.
- Esponda, I., Vespa, E., & Yuksel, S. (2024). Mental Models and Learning: The Case of Base-Rate Neglect. *American Economic Review*, 114(3), 752–782.
- Farina, A., & Herman, C. (2025). Communicating with Data Generating Processes: An Experimental Analysis.
- Feigenberg, B., & Miller, C. (2025, March). *Class Disparities and Discrimination in Traffic Stops and Searches* (Working Paper No. 33629). National Bureau of Economic Research.
- Ferrazares, T. (2025). Shift Structure and Cognitive Depletion: Evidence from Police Officers.
- Gonçalves, F., Jacome, E., & Weisburst, E. (2025). Community Engagement and Public Safety: Evidence from Crime Enforcement Targeting Immigrants.
- Gonçalves, F., & Mello, S. (2021). A Few Bad Apples? Racial Bias in Policing. *American Economic Review*, 111(5), 1406–1441.
- Gonçalves, F., Mello, S., & Weisburst, E. K. (2025, August). *Selection Bias and Racial Disparities in Police Use of Force* (tech. rep. No. w34175). National Bureau of Economic Research.
- Hoekstra, M., & Sloan, C. (2022). Does Race Matter for Police Use of Force? Evidence from 911 Calls. *American Economic Review*, 112(3), 827–860.
- Hoffman, M., Kahn, L. B., & Li, D. (2018). Discretion in Hiring. *The Quarterly Journal of Economics*, 133(2), 765–800.
- Holz, J. E., Rivera, R. G., & Ba, B. (2023). Peer Effects in Police Use of Force. *American Economic Journal: Economic Policy*, 15(2), 256–291.
- Hübert, R., & Little, A. T. (2023). A Behavioural Theory of Discrimination in Policing. *The Economic Journal*, 133(655), 2828–2843.
- Jin, G. Z., Luca, M., & Martin, D. (2021). Is no news (perceived as) bad news? An experimental investigation of information disclosure. *American Economic Journal: Microeconomics*, 13(2), 141–173.
- Kling, J. R., Liebman, J. B., & Katz, L. F. (2007). Experimental Analysis of Neighborhood Effects. *Econometrica*, 75(1), 83–119.
- Knox, D., Lowe, W., & Mummolo, J. (2020). Administrative Records Mask Racially Biased Policing. *American Political Science Review*, 114(3), 619–637.
- Koehler, J. J., & Mercer, M. (2009). Selection Neglect in Mutual Fund Advertisements. *Management Science*, 55(7), 1107–1121.
- Lum, K., & Isaac, W. (2016). To Predict and Serve? *Significance*, 13(5), 14–19.
- Maćkowiak, B., Matějka, F., & Wiederholt, M. (2023). Rational Inattention: A Review. *Journal of Economic Literature*, 61(1), 226–273.
- Malladi, S., Martinez Marquina, A., & Morozov, I. (2025, March). Space Exploration.
- Malmendier, U., & Shanthikumar, D. (2007). Are small investors naive about incentives? *Journal of Financial Economics*, 85(2), 457–489.
- National Academies of Sciences, Engineering, and Medicine. (2023, May). *Reducing Racial Inequality in Crime and Justice: Science, Practice, and Policy*. National Academies Press.
- Owens, E., Weisburd, D., Amendola, K. L., & Alpert, G. P. (2018). Can You Build a Better Cop? *Criminology & Public Policy*, 17(1), 41–87.
- Pierson, E., Simoiu, C., Overgoor, J., Corbett-Davies, S., Jenson, D., Shoemaker, A., Ramachandran, V., Barghouty, P., Phillips, C., Shroff, R., & Goel, S. (2020). A large-scale analysis of racial disparities in police stops across the United States. *Nature Human Behaviour*, 4(7), 736–745.
- Reutskaja, E., Nagel, R., Camerer, C. F., & Rangel, A. (2011). Search Dynamics in Consumer Choice under Time Pressure: An Eye- Tracking Study. *The American Economic Review*, 101(2), 900–926.
- Schwartzstein, J. (2014). Selective Attention and Learning. *Journal of the European Economic Association*, 12(6), 1423–1452.
- Stevenson, M. T., & Doleac, J. L. (2024). Algorithmic Risk Assessment in the Hands of Humans. *American Economic Journal: Economic Policy*, 16(4), 382–414.

Vomfell, L., & Stewart, N. (2021). Officer bias, over-patrolling and ethnic disparities in stop and search. *Nature Human Behaviour*, 5(5), 566–575.

A Appendix

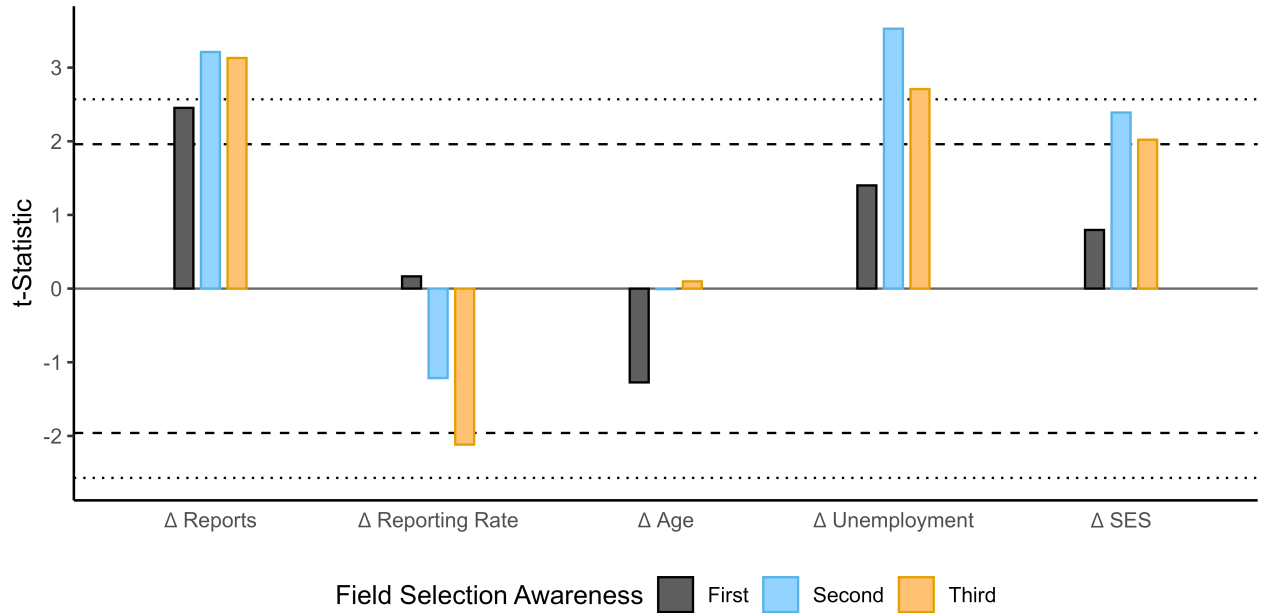
Additional Figures and Tables

Figure A.1: Prediction Bias Across Groups



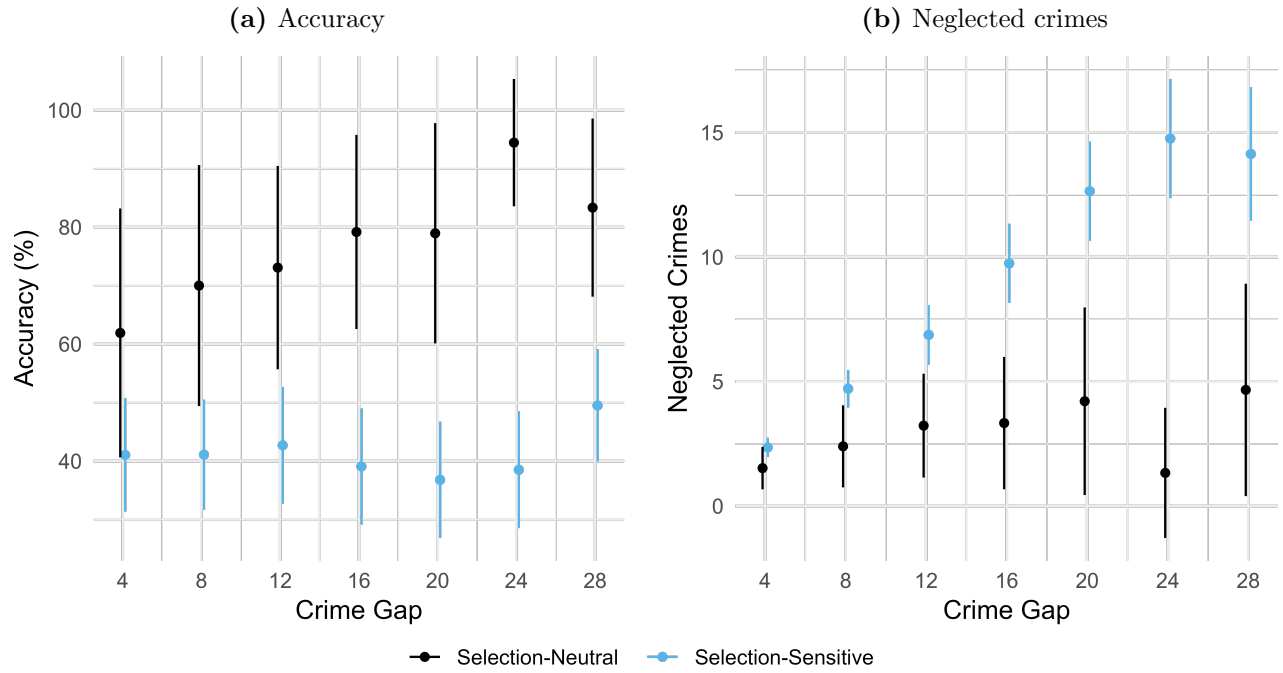
Notes: This figure shows prediction bias across groups defined as the deviation from the optimal likelihood to predict higher crime in one of the groups. We first calculate the officer-level optimal benchmark: how often their optimal prediction would be city C. Note that, although on expectation both cities have equal crime, because of randomization the benchmark is not 50% for all officers. We then calculate prediction bias as the deviation from this optimal benchmark, in absolute value, for each officer. Panel (a) shows the prediction bias across Selection Bias conditions. The left column shows the average officer in the *Uniform Selection* treatment, where reporting rates for both cities are equal in expectation. The right column shows officers in the *Selection Bias Across Groups* treatment, where reporting rates are 30 percentage points higher in expectation for one city. The figure only includes the first six predictions of Part II. Panel (b) shows the prediction bias across Endogenous Selection conditions. The left column shows the average officer in the *Exogenous Selection* group, where crime predictions don't affect reporting rates. The right column shows officers in the *Endogenous Selection* treatment, where predicting higher crime in one city increases reporting rates by 15 percentage points in that same city for the next decision. The panel includes the last six predictions of Part II, and only includes officers randomized to see no algorithmic recommendation. In both panels, error bars show the 95% confidence interval of the means, with standard errors clustered by officer. The annotations show the difference between conditions, alongside with the p-value from a t-test clustered at the officer level. The horizontal black line at zero shows the optimal benchmark: because both cities have the same number of crimes in expectation, optimal predictions should be unbiased across cities.

Figure A.2: Predictors of crime predictions, by terciles of field selection awareness



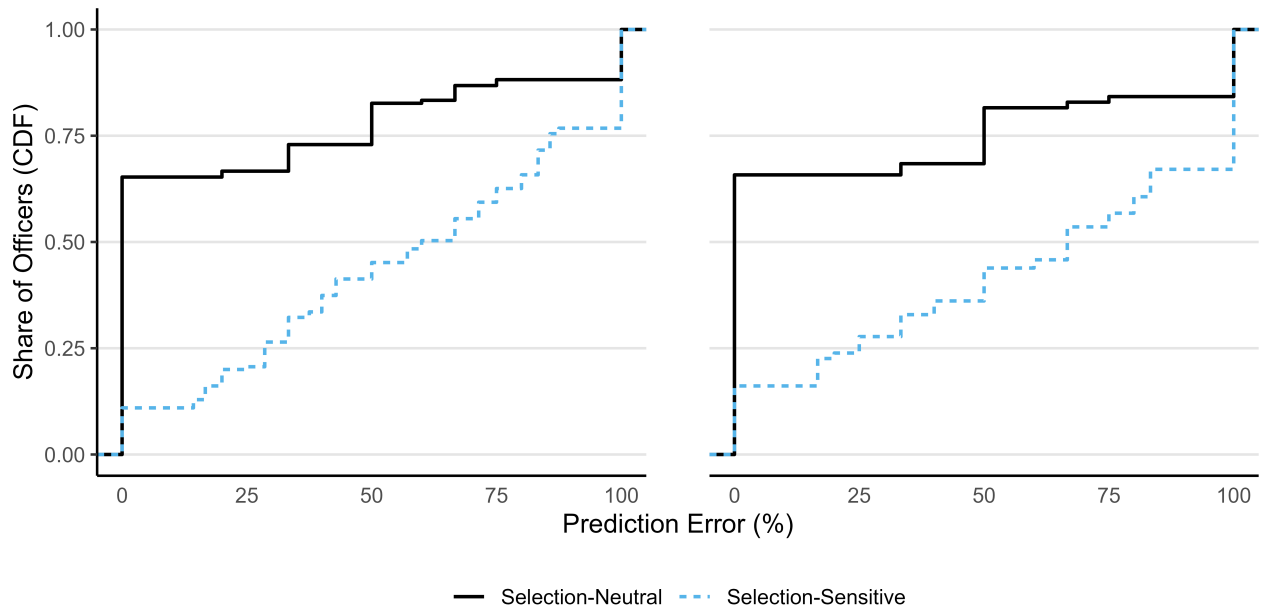
Notes: This figure shows the results of regressing crime predictions on the differences between available variables in Part I, estimated separately by terciles of Field Selection Awareness. The x-axis shows the dependent variables: Δ Reports is the difference in crime reports between the two presented neighborhoods at each prediction; Δ Reporting Rate is the difference in reporting rates, in percentage points; Δ Age is the difference in mean age; Δ Unemployment is the difference in unemployment rates, in percentage points; Δ SES is the difference in socioeconomic status (*estrato*, measured from 1 to 6). The y-axis shows the t-statistic of that regressor, estimated separately by tercile of Field Selection Awareness. We use t-statistics to ease comparison between regressors and models. Field Selection Awareness is a composite index measuring how aware officers are about selection bias in the reported crime data they work with in the field. Horizontal dashed lines mark the conventional significance level of 95%, the dotted lines mark the 99% level.

Figure A.3: Accuracy and neglected crimes by crime gap and prediction type



Notes: This figure disaggregates Figure 3 by crime gap, this is, the difference in crimes between neighborhoods at each prediction. Panel (a) shows the average prediction accuracy, by crime gap. Panel (b) shows the average number of neglected crimes —the difference in crimes between the optimal prediction and the actual one, by crime gap. Neglected crimes can be interpreted as the crime gap multiplied by the inaccuracy rate at that gap. Both panels further divide the data by whether adjusting for selection was needed to make the accurate prediction. Selection-neutral decisions, where adjusting for selection doesn't change the optimal prediction, are depicted in black; selection-sensitive decisions, where adjusting for selection does change the optimal prediction, are depicted in blue. Error bars in both panels represent the 95% confidence interval of the mean, with standard errors clustered at the officer level.

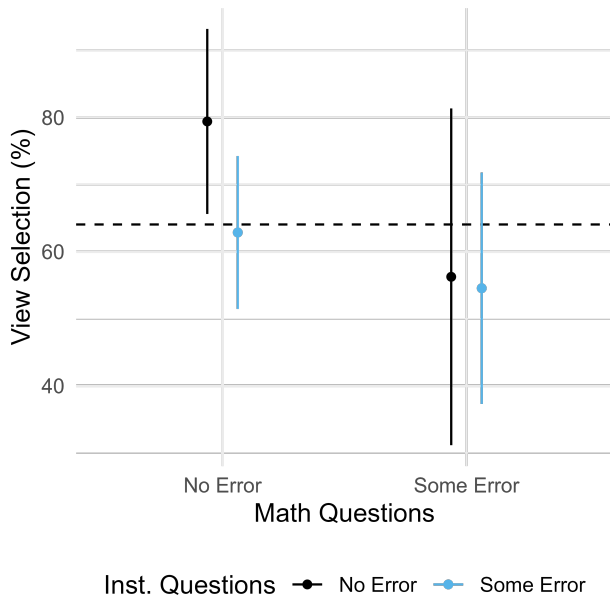
Figure A.4: Accuracy and neglected crimes by crime gap and prediction type



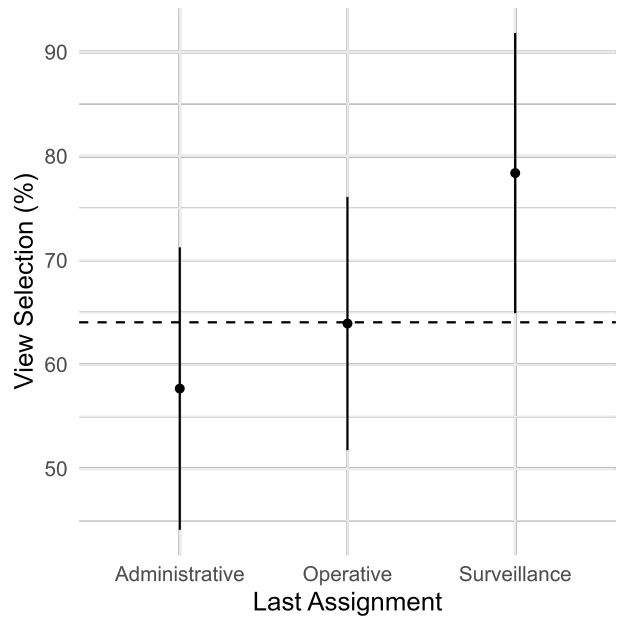
Notes: This figure shows the cumulative distribution of officer-level prediction error by type of prediction. Both panels divide the data by whether adjusting for selection was needed to make the right prediction. Selection-neutral decisions, where adjusting for selection doesn't change the optimal prediction, are depicted in black; selection-sensitive decisions, where adjusting for selection does change the optimal prediction, are in blue dashed lines. The left panel uses all eight predictions from Part I, the right panel uses the first six predictions from Part II.

Figure A.5: Choice to View Selection

(a) By performance in math and instructions questions



(b) By last assignment



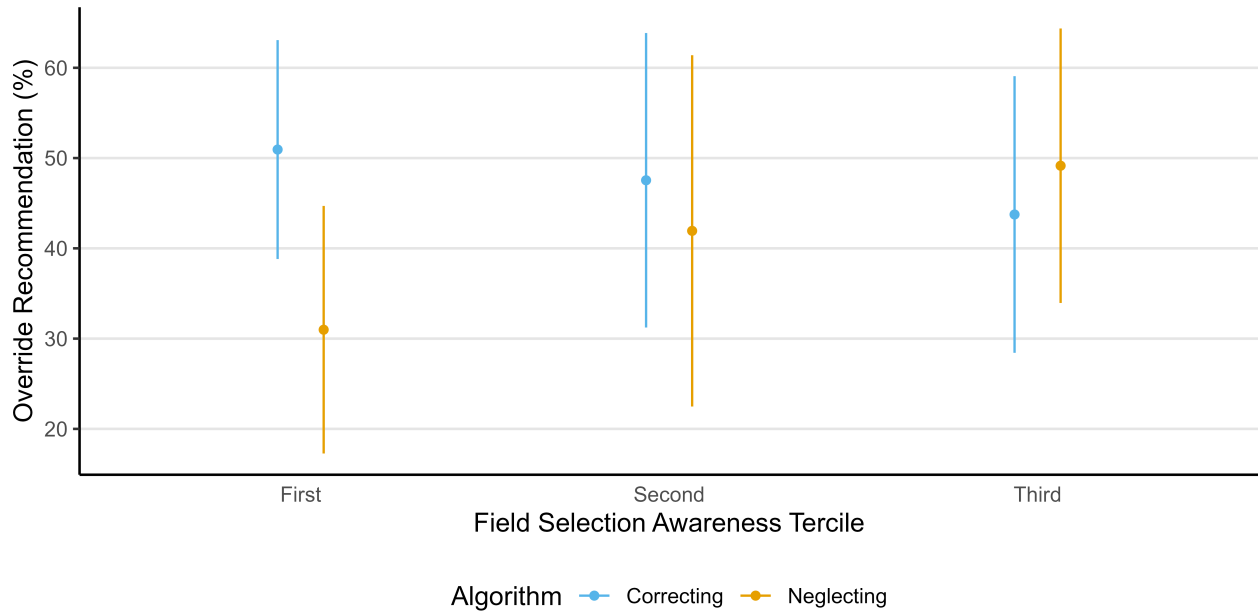
Notes: This figure shows the share of officers who choose to view selection for the last prediction of Part I. Panel (a) divides the sample by performance in the math and instruction comprehension questions. The left column shows officers who answered all math questions correctly, while the right column shows those who made at least one error. In black, we show officers who answered all instruction comprehension questions correctly, while in black we show those who made at least one error. Panel (b) divides the sample by the last assignment of the officer before entering the academy. Note that those with a Vigilance assignment were in charge of predicting crime and assigning patrols. In both panels, error bars represent the 95% confidence interval around the mean. The dashed line shows the sample-level mean at 64%.

Figure A.6: Accuracy Distributions in Selection-Sensitive Decisions



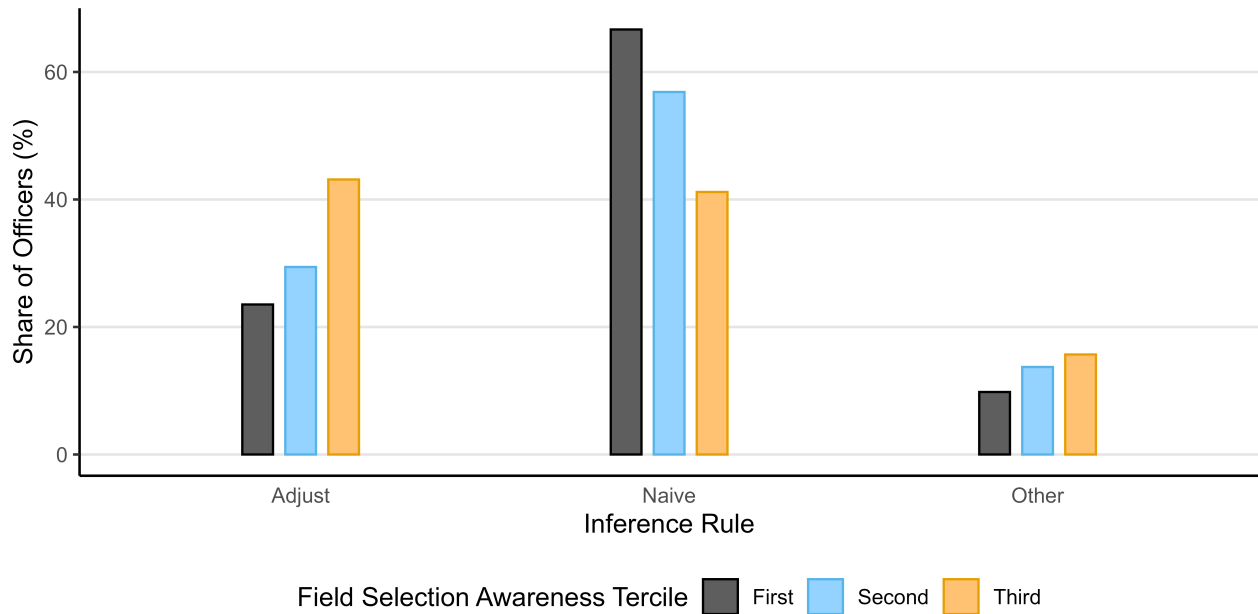
Notes: This figure shows the distribution of officer-level prediction accuracy in selection-sensitive decisions. Plots in the first row show predictions from Part I, where data selection is less salient. Plots in the second row show prediction from Part II, where data selection is made more salient. Columns separate the sample between officers who chose *not* to view the reporting rate for the last prediction of Part I (left, in black), and officers who chose to view selection (right, in blue). When comparing across columns, we notice the strong link between choosing to see selection and prediction accuracy in Part I, where selection wasn't salient. Once selection is made more salient in Part II, officers who hadn't chosen to view it in Part I (left column) don't really improve accuracy but become more likely to systematically neglect it. For officers who already were paying attention to reporting rates, the increase in salience doesn't seem to have any effect. In both cases, salience doesn't mitigate neglect. While bottom-up salience increases the likelihood that selection comes to mind, it does not substitute for having the correct adjustment rule.

Figure A.7: Heterogeneity in Advice Override



Notes: This figure shows the share of selection-sensitive decisions in which officers override the algorithm recommendation, dividing the sample in terciles according to their field selection awareness. Error bars show the 95% confidence interval of the mean, with standard errors clustered at the officer level.

Figure A.8: Inference Rules



Notes: This figure shows the share of selection-sensitive decisions in which officers override the algorithm recommendation, dividing the sample in terciles according to their field selection awareness. Error bars show the 95% confidence interval of the mean, with standard errors clustered at the officer level.

Table A.1: Prediction error across decisions

	Prediction Error					
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.221 (0.000)	0.327 (0.000)	0.159 (0.000)	0.230 (0.000)	0.333 (0.000)	0.136 (0.006)
Selection-Sensitive	0.358 (0.000)	0.282 (0.000)	0.495 (0.000)	0.362 (0.000)	0.323 (0.003)	0.509 (0.000)
View Selection		-0.165 (0.007)			-0.155 (0.111)	
Selection-Sensitive $\times$ View Selection		0.119 (0.183)			0.047 (0.704)	
Calculator			0.011 (0.867)			0.103 (0.310)
Selection-Sensitive $\times$ Calculator			-0.299 (0.002)			-0.427 (0.002)
Observations	1,240	1,224	784	930	918	588
Dependent variable mean	49.274	49.346	46.811	53.333	52.832	48.469

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable Prediction Error is a dummy that equals 1 if the prediction of which neighborhood had more crime was wrong. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. View Selection is a dummy that equals 1 if the officer chose to view the reporting rate for the last decision of Part I. Calculator is a dummy that equals 1 if that officer ever used the on-screen calculator through that part. Columns (1) to (3) use the eight predictions from Part I, with Column (3) restricting the sample to those who chose to view selection in the last decision of Part I. Columns (4) to (6) use all predictions in Part II, excluding the last six predictions for participants randomized into an algorithm treatment other than *Control*. Column (6) restricts the sample to those who chose to view selection in the last decision of Part I.

Table A.2: Neglected crimes across decisions

	Neglected Crimes			
	Part I		Part II	
	(1)	(2)	(3)	(4)
Constant	9.880 (0.000)	10.466 (0.000)	3.026 (0.000)	1.741 (0.034)
Selection-Sensitive	11.273 (0.000)	-8.718 (0.001)	6.367 (0.000)	-0.997 (0.285)
Crime Gap		-0.011 (0.799)		0.081 (0.228)
Selection-Sensitive $\times$ Crime Gap		0.516 (0.000)		0.459 (0.000)
Observations	1,240	1,240	930	930
Dependent variable mean	18.435	18.435	8.353	8.353

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable Neglected Crimes measures the prediction error in each decision: the difference between the highest number of crimes across the two neighborhoods and the predicted neighborhood. Note that this variable takes a value of zero if the prediction was correct, and the crime gap between neighborhoods if the prediction was wrong. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Crime Gap measures the difference in crimes between the two neighborhoods. Columns (1) and (2) use the eight predictions from Part I. Columns (3) and (4) use all predictions in Part II, excluding the last six predictions for participants randomized into an algorithm treatment other than *Control*.

Table A.3: Effect of crime and reporting gaps in prediction probability

	Predict A			Predict C		
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.469 (0.000)			0.499 (0.000)		
Crime Gap	0.002 (0.000)	0.003 (0.000)	0.001 (0.098)	0.003 (0.011)	0.003 (0.007)	-0.001 (0.674)
Reporting Gap	0.003 (0.000)	0.004 (0.000)	0.003 (0.000)	0.003 (0.000)	0.003 (0.000)	0.003 (0.008)
View Selection $\times$ Crime Gap			0.002 (0.062)			0.006 (0.009)
View Selection $\times$ Reporting Gap			0.000 (0.700)			0.000 (0.980)
<i>Fixed-effects</i>						
Officer	No	Yes	Yes	No	Yes	Yes
Observations	1,224	1,224	1,224	918	918	918
Dependent variable mean	0.469	0.469	0.469	0.487	0.487	0.487

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable Predict A is a dummy that equals 1 if the neighborhood from city A was predicted to have higher crime (Columns (1) to (3)). The dependent variable Predict C is a dummy that equals 1 if the neighborhood from city C was predicted to have higher crime, using predictions from Part II (Columns (4) to (6)). Crime Gap is the difference in crimes between the two presented neighborhoods at each prediction. Reporting Gap is the difference in reporting rates between the two presented neighborhoods at each prediction, in percentage points. View Selection is a dummy that equals 1 if the officer chose to view the reporting rate for the last prediction of Part I. Columns (1) to (3) use the eight predictions of Part I, including the gap in demographic decoys as controls. Columns (4) to (6) use the first six predictions in Part II. Columns (2), (3), (5), and (6) include officer fixed effects.

Table A.4: Effect of all observable variables in prediction probability

	Predict A			Predict C		
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.467 (0.000)			0.494 (0.000)		
Δ Crime Reports	0.003 (0.000)	0.005 (0.000)	0.003 (0.034)	0.003 (0.026)	0.004 (0.002)	0.000 (0.950)
Reporting Gap	-0.003 (0.088)	-0.006 (0.002)	-0.003 (0.325)	-0.005 (0.178)	-0.008 (0.026)	0.004 (0.434)
Δ Age	-0.001 (0.422)	-0.001 (0.531)	-0.003 (0.279)			
Δ Unempt.	0.010 (0.000)	0.008 (0.001)	0.021 (0.000)			
Δ SES	0.022 (0.005)	0.017 (0.032)	0.037 (0.012)			
View Selection $\times$ Δ Crime Reports			0.003 (0.105)			0.007 (0.008)
View Selection $\times$ Reporting Gap			-0.006 (0.159)			-0.018 (0.008)
View Selection $\times$ Δ Age			0.003 (0.472)			
View Selection $\times$ Δ Unempt.			-0.020 (0.000)			
View Selection $\times$ Δ SES			-0.031 (0.067)			
<i>Fixed-effects</i>						
Officer	No	Yes	Yes	No	Yes	Yes
Observations	1,224	1,224	1,224	918	918	918
Dependent variable mean	0.469	0.469	0.469	0.487	0.487	0.487

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable Predict A is a dummy that equals 1 if the neighborhood from city A was predicted to have higher crime (Columns (1) to (3)). The dependent variable Predict C is a dummy that equals 1 if the neighborhood from city C was predicted to have higher crime, using predictions from Part II (Columns (4) to (6)). Δ Crime Reports is the difference in crime reports between the two presented neighborhoods at each prediction. Reporting Gap is the difference in reporting rates between the two presented neighborhoods at each prediction, in percentage points. Δ Age is the difference in mean age between the two presented neighborhoods at each prediction. Δ Unemployment is the difference in unemployment rates between the two presented neighborhoods at each prediction, in percentage points. Δ SES is the difference in socioeconomic status (*estrato*, measured from 1 to 6) between the two presented neighborhoods. View Selection is a dummy that equals 1 if the officer chose to view the reporting rate for the last prediction of Part I. Columns (1) to (3) use the eight predictions of Part I. Columns (4) to (6) use the first six predictions in Part II. Columns (2), (3), (5), and (6) include officer fixed effects.

Table A.5: Accuracy and neglected crimes by selection, robustness

	Prediction Error				Missed Crimes			
	Part I		Part II		Part I		Part II	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	0.223 (0.000)	0.177 (0.000)	0.240 (0.000)	0.311 (0.000)	9.637 (0.000)	7.956 (0.000)	3.042 (0.000)	4.089 (0.004)
Selection-Sensitive	0.291 (0.000)	0.328 (0.000)	0.304 (0.000)	0.226 (0.030)	8.781 (0.000)	10.424 (0.001)	5.504 (0.000)	4.570 (0.009)
Some Math Error	-0.006 (0.921)		-0.025 (0.790)		0.683 (0.800)		-0.042 (0.978)	
Selection-Sensitive $\times$ Some Math Error	0.211 (0.020)		0.178 (0.166)		7.839 (0.052)		2.680 (0.195)	
Some Inst. Error		0.070 (0.221)		-0.115 (0.248)		3.093 (0.236)		-1.509 (0.346)
Selection-Sensitive $\times$ Some Inst. Error		0.036 (0.685)		0.197 (0.114)		0.898 (0.816)		2.602 (0.204)
Observations	1,240	1,240	930	930	1,240	1,240	930	930
Dependent variable mean	0.493	0.493	0.533	0.533	18.435	18.435	8.353	8.353

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable Prediction Error is a dummy that equals 1 if the prediction of which neighborhood had more crime was wrong (Columns (1) to (4)). The dependent variable Missed Crimes measures the prediction error in each decision: the difference between the highest number of crimes across the two neighborhoods and the predicted neighborhood (Columns (5) to (8)). Note that this variable takes a value of zero if the prediction was correct, and the crime gap between neighborhoods if the prediction was wrong. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Some Math Error is a dummy that equals 1 if the officer didn't answer all math questions correctly. Some Instruction Error is a dummy that equals 1 if the officer didn't answer all instruction comprehension questions correctly on the first attempt. Columns (1), (2), (5), and (6) use the eight predictions from Part I. Columns (3), (4), (7), and (8) use the first six predictions in Part II.

Table A.6: Heterogeneity in choice to view selection

	View Selection			
	(1)	(2)	(3)	(4)
Constant	0.762 (0.000)	0.577 (0.000)	0.649 (0.000)	0.708 (0.000)
Some Math Error	-0.132 (0.125)			-0.136 (0.107)
Some Instruction Error	-0.118 (0.139)			-0.135 (0.087)
Operative		0.062 (0.503)		0.076 (0.412)
Surveillance		0.207 (0.035)		0.193 (0.051)
Reported Data Reliability			0.107 (0.212)	0.110 (0.181)
Observations	153	150	151	150
Dependent variable mean	0.641	0.653	0.649	0.653

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable View Selection is a dummy that equals 1 if the officer chose to view the reporting rates for the tenth decision of Part I. Some Math Error is a dummy that equals 1 if the officer didn't answer all math questions correctly. Some Instruction Error is a dummy that equals 1 if the officer didn't answer all instruction comprehension questions correctly on the first attempt. Operative is a dummy that equals 1 if the officer's last assignment was operative. Surveillance is a dummy that equals 1 if the officer's last assignment was in surveillance —predicting crime and assigning patrols. The reference group is officers whose last assignment was administrative. Reported Data Reliability measures the officer's perception of how reliably reported crime data captures underlying crime, with higher values implying higher reliability. All columns use data at the officer level, as there is only one choice to view selection.

Table A.7: Heterogeneity in accuracy loss due to selection

	Accuracy					
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.788 (0.000)	0.782 (0.000)	0.788 (0.000)	0.732 (0.000)	0.732 (0.000)	0.754 (0.000)
Selection-Sensitive	-0.428 (0.000)	-0.358 (0.000)	-0.426 (0.000)	-0.378 (0.001)	-0.338 (0.000)	-0.400 (0.000)
Operative	0.002 (0.971)		0.007 (0.917)	-0.009 (0.921)		-0.030 (0.688)
Surveillance	-0.038 (0.609)		-0.050 (0.465)	0.004 (0.972)		-0.034 (0.706)
Selection-Sensitive $\times$ Operative	0.131 (0.168)		0.118 (0.208)	0.102 (0.433)		0.120 (0.283)
Selection-Sensitive $\times$ Surveillance	0.082 (0.485)		0.104 (0.343)	0.015 (0.915)		0.064 (0.611)
Report Reliability		0.115 (0.082)	0.124 (0.068)		0.278 (0.000)	0.281 (0.000)
Selection-Sensitive $\times$ Report Reliability		-0.255 (0.010)	-0.262 (0.008)		-0.394 (0.000)	-0.406 (0.000)
Observations	1,216	1,224	1,216	1,224	1,236	1,224
Dependent variable mean	0.512	0.511	0.512	0.467	0.462	0.467

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable Accuracy is a dummy that equals 1 if the prediction of which neighborhood had more crime was accurate. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Operative is a dummy that equals 1 if the officer's last assignment was operative. Surveillance is a dummy that equals 1 if the officer's last assignment was in surveillance —predicting crime and assigning patrols. The reference group is officers whose last assignment was administrative. Report Reliability (Reported Data Reliability) measures the officer's perception of how reliably reported crime data captures underlying crime, with higher values implying higher reliability. Columns (1) to (3) use the eight predictions from Part I. Columns (4) to (6) use all predictions in Part II, excluding the last six predictions for participants randomized into an algorithm treatment other than *Control*.

Table A.8: Selection Neglect across Selection Environments

	Accuracy		
	(1)	(2)	(3)
Constant	1.000 (0)	0.565 (0.000)	1.000 (0.000)
Selection-Sensitive	-0.591 (0.000)	-0.167 (0.260)	-0.600 (0.000)
Asymmetric	-0.235 (0.000)		-0.455 (0.000)
Selection-Sensitive $\times$ Asymmetric	0.230 (0.006)		0.449 (0.034)
Endogenous		0.132 (0.324)	-0.345 (0.003)
Selection-Sensitive $\times$ Endogenous		-0.214 (0.236)	0.236 (0.146)
Endogenous $\times$ Asymmetric			0.523 (0.006)
Selection-Sensitive $\times$ Endogenous $\times$ Asymmetric			-0.460 (0.127)
Observations	930	330	330
Dependent variable mean	0.467	0.455	0.455

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable Accuracy is a dummy that equals 1 if the prediction of which neighborhood had more crime was accurate. Asymmetric is a dummy variable that equals 1 if the officer was randomized into the *Asymmetric Selection* treatment, where neighborhoods from one of the two cities have a 30 percentage points higher reporting rate in expectation than the other city. It equals 0 when the officer is in the *Symmetric Selection* treatment, where reporting rates for both cities are drawn from the same uniform distribution, and thus are equal in expectation. Endogenous is a dummy variable that equals 1 if an officer is randomized into the *Endogenous Selection* treatment, where predicting one city to have higher crime increases reporting rates in that city by 15 percentage points in the next prediction. It equals 0 when the officer is in the *Exogenous Selection* treatment, where predictions don't affect future reporting rates. Column (1) uses the first six predictions of Part II. Columns (2) and (3) use the last six predictions in Part II, excluding officers randomized into an algorithm treatment other than *Control*.

Table A.9: Effects of selection environments

	Chose High Rate	Repeat Prediction	Persistent Prediction	
	(1)	(2)	(3)	(4)
Constant	0.496 (0.000)	0.532 (0.000)	0.056 (0.153)	0.000 (1.000)
Systematic Bias	0.104 (0.006)		0.150 (0.063)	0.200 (0.134)
Endogenous		0.146 (0.028)	0.201 (0.014)	0.214 (0.065)
Systematic Bias $\times$ Endogenous			-0.037 (0.774)	-0.148 (0.486)
Observations	930	330	155	55
Dependent variable mean	0.549	0.609	0.232	0.164

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable Chose High Rate is a dummy variable that equals 1 if the city with the higher average reporting rate was predicted to have higher crime. The dependent variable Repeat Prediction is a dummy that equals 1 if the city predicted to have higher crime is the same as in the previous prediction. The dependent variable Persistent Prediction is a dummy that equals 1 if an officer predicted the same city to have higher crime across the last six decisions of Part II. Systematic Bias is a dummy variable that equals 1 if the officer was randomized into the *Selection Bias Across Groups* treatment, where neighborhoods from one of the two cities have a 30 percentage points higher reporting rate in expectation than the other city. It equals 0 when the officer is in the *Uniform Selection Bias* treatment, where reporting rates for both cities are drawn from the same uniform distribution, and thus are equal in expectation. Endogenous is a dummy variable that equals 1 if an officer is randomized into the *Endogenous Selection* treatment, where predicting one city to have higher crime increases reporting rates in that city by 15 percentage points in the next prediction. It equals 0 when the officer is in the *Exogenous Selection* treatment, where predictions don't affect future reporting rates. Column (1) uses the first six predictions of Part II. Column (2) uses the last six predictions in Part II, only including officers randomized into the Control algorithm.. Column (3) aggregates the last six predictions of Part II at the officer level. Column (4) repeats this analysis but only including officers randomized into the Control algorithm.

Table A.10: Prediction Bias Across Groups, Selection Environments

	Prediction Bias		
	(1)	(2)	(3)
Constant	0.178 (0.000)	0.218 (0.000)	0.208 (0.000)
Selection Bias	0.093 (0.006)		0.025 (0.734)
Endogenous		0.104 (0.059)	0.149 (0.028)
Endogenous $\times$ Selection Bias			-0.093 (0.395)
Observations	155	55	55
Dependent variable mean	0.226	0.273	0.273

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable Prediction Bias is the officer-level deviation from the optimal likelihood to predict higher crime in one of the groups. We first calculate the officer-level optimal benchmark: how often their optimal prediction would be city C. Note that, although on expectation both cities have equal crime, because of randomization the benchmark is not 50% for all officers. We then calculate prediction bias as the deviation from this optimal benchmark, in absolute value, for each officer. Selection Bias is a dummy variable that equals 1 if the officer was randomized into the *Selection Bias Across Groups* treatment, where neighborhoods from one of the two cities have a 30 percentage points higher reporting rate in expectation than the other city. It equals 0 when the officer is in the *Uniform Selection Bias* treatment, where reporting rates for both cities are drawn from the same uniform distribution, and thus are equal in expectation. Endogenous is a dummy variable that equals 1 if an officer is randomized into the *Endogenous Selection* treatment, where predicting one city to have higher crime increases reporting rates in that city by 15 percentage points in the next prediction. It equals 0 when the officer is in the *Exogenous Selection* treatment, where predictions don't affect future reporting rates. Column (1) uses the first six predictions of Part II. Column (2) and (3) use the last six predictions in Part II, only including officers randomized into the Control algorithm.

Table A.11: Standardized response times by decision features

	Standardized Response Time					
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	-0.085 (0.167)	0.272 (0.038)	0.192 (0.114)	-0.114 (0.119)	0.298 (0.135)	0.071 (0.683)
Accurate	0.167 (0.031)	-0.411 (0.003)	-0.481 (0.000)	0.245 (0.004)	-0.468 (0.025)	-0.434 (0.021)
Selection-Sensitive		-0.400 (0.003)	-0.365 (0.003)		-0.444 (0.032)	-0.317 (0.080)
Selection-Sensitive $\times$ Accurate		0.752 (0.000)	0.713 (0.000)		0.855 (0.001)	0.566 (0.009)
Use Calculator			1.747 (0.000)			1.323 (0.000)
Observations	1,240	1,240	1,240	930	930	930
Dependent variable mean	0.000	0.000	0.000	0.000	0.000	0.000

Notes: Results from a linear regression with p -values in parentheses and standard errors clustered at the officer level. The dependent variable Standardized Response Time is the z-scored response time for each prediction. Accurate is a dummy that equals 1 if the prediction of which neighborhood had more crime was accurate. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Use Calculator is a dummy that equals 1 if the on-screen calculator was used for that prediction. Columns (1) to (3) use the eight predictions from Part I. Columns (4) to (6) use the first six predictions of Part II.